

A Correlation-Aware Graph Transformer Framework for Joint Regional Load and Operational Stress Forecasting in Bangladesh

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Abstract—Day-ahead planning in developing-economy national grids requires synchronised visibility over regional electricity demand and system-wide operational pressure, yet many pipelines treat load prediction and stress monitoring separately. On the Bangladesh national power system, daily records across nine administrative divisions co-report evening-peak load with limitation, reserve, and shedding indicators, motivating coupled rather than isolated forecasting. This study investigates whether a spatio-temporal model can jointly forecast next-day regional demand and a graph-level Operational Stress Index from shared historical inputs under chronological evaluation. The proposed Correlation-Aware Multi-Task Forecasting Framework is implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The model encodes division-level and national context over a fixed lookback window, conditions representations on a train-estimated correlation graph, and decodes regional demand and a bounded system-wide stress scalar through shared spatio-temporal encoding and task-specific heads, with a repaired multi-task loss protocol that stabilises joint learning. Evaluation on a frozen chronologically held-out test year comprises classical, recurrent, and graph-convolution benchmarks; graph and architecture ablations; regional error analysis; and multi-method post-hoc attribution. Within the evaluated Bangladesh setting, the selected correlation-graph configuration attains the strongest joint demand-and-stress credentials in the internal suite, with statistically supported demand improvement over the historical hybrid reference and comparatively informative OSI outputs relative to stress-capable baselines. Ablation evidence favours correlation adjacency over geographical or hybrid priors on this dataset, while documenting a trade-off between demand-only accuracy and joint stress forecasting; attribution analysis highlights limitation and calendar coalitions and reports partial cross-method agreement. The work contributes a reproducible national-case study of graph-transformer multi-task forecasting for regional load and operational stress in Bangladesh, scoped to offline evaluation rather than universal superiority or demonstrated deployment outcomes.

Index Terms—Multi-task learning, Spatio-temporal forecasting, Operational stress index, Load forecasting, Correlation graph, Smart grid, Explainability, Bangladesh power grid

I. INTRODUCTION

Electricity system operators in national grids must anticipate regional load and assess system pressure on a day-ahead horizon to coordinate generation dispatch, reserve planning, and constraint management. In developing economies, where supply limitations, infrastructure stress, and intermittent load shedding can coincide with rapid demand growth, planning

visibility depends on timely forecasts that reflect both where load will concentrate across administrative regions and how severely operational constraints may bind at the system level. Unlike highly instrumented transmission networks with dense phasor measurement coverage, many national systems in South Asia report operational state through daily aggregated division-level balances, national generation scalars, and qualitative limitation indicators that nevertheless encode recurring patterns of stress when reserves tighten or fuel and infrastructure constraints bind. Historical operational records from smart-grid reporting platforms increasingly offer daily traces of division-level demand, supply shortfalls, fuel and maintenance limitations, and national generation context. These records create an opportunity to move beyond aggregate national load models toward spatio-temporal learning approaches that respect inter-regional coupling while incorporating system-wide stress signals relevant to control-centre review. For the Bangladesh national power system, daily operational data are reported across nine administrative divisions over a multi-year timeline, providing a documented case in which regional evening-peak demand and limitation dynamics are co-visible within each daily record [1]. Accurate short-horizon forecasting in such settings is not merely a statistical exercise: planners require forecasts that are spatially disaggregated, temporally aligned to the operational reporting cadence, and interpretable enough to support human review when stress indicators and load trajectories must be judged together.

Despite the availability of rich operational timelines, forecasting practice in power systems often treats load prediction and stress monitoring as separate analytical threads. Classical and ensemble machine-learning models [2]–[4] typically forecast demand at regional or national scale without an explicit graph structure to encode coupling among divisions, while shallow recurrent or graph-convolution architectures [5], [6] may struggle to combine long-range temporal context with informative multi-node representations on national footprints. Graph-based methods [7]–[9] imported from residential, microgrid, or infrastructure-topology settings frequently rely on geographical adjacency or fixed network maps that may not capture data-driven co-movement among regional loads in developing-economy grids. At the same time, operational stress—arising from reserve shortfall, aggregated limitation

pressure, and shedding intensity—is seldom forecast jointly with regional demand in a single multi-task model; shedding-oriented studies in the literature [10], [11] more often address control, optimisation, or under-frequency response than day-ahead forecasting of graded stress indices for planning visibility. Explainability, where present [12], [13], is rarely integrated with multi-node graph-temporal forecasts that emit coupled load and stress outputs for operator-facing review. The result is a persistent mismatch between what operational records contain—synchronised regional load and system constraint context—and what many forecasting pipelines deliver: disconnected single-task predictions with limited spatial structure and limited transparency.

The present study addresses that mismatch by formulating day-ahead forecasting as a supervised multi-task problem [14] on a fixed set of spatial units. The primary task is next-day regional evening-peak demand across nine divisions; the secondary task is a graph-level Operational Stress Index (OSI) that summarises system-wide pressure from shedding intensity, generation reserve margin, and operational limitation components normalised under a leakage-safe training policy. OSI is constructed as a bounded composite score rather than a sparse per-division shedding label, preserving graded stress magnitude on days when reserve or limitation pressure is elevated even if shedding is absent. The two targets are operationally related yet statistically non-redundant: demand describes expected load levels by region, whereas OSI encodes constraint-related pressure that is not fully determined by demand alone. Joint estimation from shared spatio-temporal inputs is motivated by the need for synchronised day-ahead visibility rather than by treating stress prediction as an incidental add-on to load modelling. The research problem therefore asks whether a single spatio-temporal deep learning framework can learn coupled regional demand and operational stress forecasts from historical Bangladesh grid records under chronological evaluation discipline, and which design choices—graph prior, architectural components, task weighting, and post-hoc attribution—are empirically justified on this national case.

A structured review of the collected literature and an evidence-based gap analysis identify several limitations that motivate the present work without duplicating a full survey here. First, peer-reviewed spatio-temporal load and shedding studies rarely use Bangladesh division-level smart-grid daily data; developing-economy national grids remain under-represented relative to US and European residential or microgrid benchmarks. Second, continuous multi-node demand forecasting is seldom coupled with operational-stress-style targets in one multi-task framework; task isolation between load and shedding persists in much of the corpus. Third, shedding research often targets control actions rather than forecast-centric formulations of graded daily stress suitable for planning. Fourth, only a minority of recent work combines graph-based spatial coupling with transformer-style temporal modelling [15]–[17] for multi-node national forecasting, and none of the collected comparators address the Bangladesh

shedding and limitation context directly. Fifth, explainable artificial intelligence appears in a small fraction of graph-temporal energy studies [12], [18] and is seldom reported with explicit multi-method discordance for coupled demand and stress outputs. Sixth, operational stress assessment in adjacent literature frequently emphasises transmission or microgrid assets rather than daily regional stress indices derived from demand–supply–limitation dynamics visible in national operational reports. Seventh, exogenous limitation covariates—fuel, water, maintenance, and calendar structure—are not consistently co-designed with spatio-temporal architectures in published load models. Eighth, weak documentation of chronological splits, baselines, and ablations in part of the conference literature complicates fair comparison of proposed methods. Together, these gaps define the scope of contribution: a reproducible, evidence-locked study on a previously unreported national case that couples graph-temporal learning, multi-task stress forecasting, and operator-oriented attribution within one frozen evaluation programme.

To respond to that scope, this paper proposes the Correlation-Aware Multi-Task Forecasting Framework, implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The framework ingests a fixed lookback window of node-level and national context features for each administrative division, together with a train-estimated correlation graph that encodes inter-regional demand coupling from training-partition data only. A parallel dual-branch encoder combines graph-aware spatial attention with temporal transformer capacity [7], [17] over the lookback window; learned fusion feeds shared representations into task-specific heads that decode regional megawatt demand and a bounded system-wide OSI scalar for the next operating day. Graph construction evaluates geographical, hybrid, and correlation-only priors; the proposed configuration adopts correlation-thresholded adjacency as the definitive spatial prior following empirical comparison rather than assuming that geographical proximity alone captures predictive coupling on this dataset. Multi-task training employs a scale-aware joint loss with explicit stress weighting [14], [19] so that shared representations encode limitation-relevant structure without collapsing to demand-only learning—a design requirement identified during protocol repair when unconstrained multi-task optimisation failed to produce informative stress outputs. Post-hoc attribution applies grouped integrated gradients, permutation importance, spatial and temporal attention export [2], [18], [20], and stratified case studies on the frozen final checkpoint to supply coalition-level, spatial, and dual-path stress explanations for operator review, with method disagreement reported explicitly rather than treated as validation of physical mechanisms. Inputs combine engineered autoregressive, calendar, limitation, and grid-aggregate features with a seven-day lookback aligned to the daily reporting cadence of the source timeline. The framework is evaluated on a chronologically partitioned timeline with locked data artefacts, internal benchmarks spanning classical, recurrent, and graph-convolution references [2], [5], [21], configuration ablations, architecture simplification, error

geography, and explainability analysis; detailed protocol and implementation specifications are deferred to later sections.

The principal contributions of this study are as follows. First, a Correlation-Aware Multi-Task Forecasting Framework (final configuration S2) is introduced for joint day-ahead regional demand and graph-level OSI prediction from a single spatio-temporal graph-transformer model on Bangladesh national grid data. Second, an evidence-driven architecture selection process replaces the initial hybrid geographical–correlation graph design with a correlation-only adjacency prior while retaining full temporal encoder capacity where ablation indicates it is consequential for the selected topology. Third, a repaired multi-task training protocol with explicit demand–stress loss balancing enables stable joint learning of load and stress targets under the frozen W20 specification. Fourth, a rigorous empirical evaluation programme—comprising external benchmarks, component ablations, inferential testing on the primary demand metric with multiple-comparison correction, and metric verification for classical baselines—supports reproducible comparison within the frozen suite. Fifth, an integrated explainability analysis on the final model supplies multi-method attribution artefacts for both forecasting tasks, including spatial and temporal views and comparison against OSI component decomposition on audited case studies. These contributions are established through offline evaluation on held-out test data; they are positioned as planning and transparency evidence rather than as proof of deployment outcomes or universal architectural superiority.

The remainder of this paper is organised as follows. Section 5 reviews related work on load forecasting, graph neural networks, transformer-based temporal modelling, multi-task learning in energy systems, and explainability for forecasting. Section 6 presents the methodology, including problem formulation, dataset description, feature engineering, graph construction, PF-STGT architecture, and training protocol. Section 7 describes the experimental setup, benchmarks, ablations, and evaluation metrics. Section 8 reports results on overall benchmark performance, regional forecasts, operational stress analysis, ablation studies, error analysis, and explainability. Section 9 discusses principal findings, comparison with existing literature, practical implications, limitations, and future research directions. Section 10 concludes the study. Supplementary tables, statistical summaries, feature and graph specifications, explainability artefact paths, and a reproducibility checklist are provided in Appendix A.

II. RELATED WORK

This section surveys prior research along four themes that bound the present study: short-horizon load forecasting, graph-based spatio-temporal modelling, multi-task learning in energy systems, and explainable artificial intelligence for forecasting. The synthesis draws on a structured critical review of fifty-five peer-reviewed and preprint studies collected for the Bangladesh smart-grid forecasting programme (2023–2026), supplemented by limitation and opportunity catalogues derived from reviewer-style analysis of that corpus. The objective

is not to enumerate individual publications, but to compare methodologies, datasets, assumptions, limitations, and unresolved challenges that collectively define the research space addressed by the Correlation-Aware Multi-Task Forecasting Framework implemented as PF-STGT.

A. Load Forecasting

Short-horizon electricity load forecasting remains the largest thematic cluster in the reviewed literature. Methods span classical statistical and machine-learning regressors through deep recurrent, convolutional, and transformer architectures applied primarily to scalar or multivariate load series. A substantial share of recent work targets residential or advanced-metering-infrastructure (AMI) consumption at sub-daily or daily resolution, often assuming dense sampling, homogeneous prosumers, or building-level aggregates rather than administratively defined regional nodes on a national transmission footprint. Ensemble tree methods, gradient boosting, and hybrid statistical–deep pipelines are frequently reported as strong tabular baselines on these datasets [2]–[4], reflecting long-standing evidence that well-tuned non-graphical models remain competitive when spatial structure is implicit in feature engineering rather than explicit in the model architecture.

Deep learning contributions in the load-forecasting cluster emphasise temporal pattern extraction: stacked recurrent encoders, convolutional front-ends, Autoformer- and transformer-style sequence models [5], [22]–[24], and multi-encoder designs that fuse textual or exogenous embeddings with numerical load history. Common assumptions include stationary seasonality within the training window, availability of long uninterrupted series, and forecast horizons aligned to market or building-management cadences that may differ from daily national operational reporting. Industrial and sector-specific case studies occasionally use public national statistics—such as industrial electricity accounts in North American datasets—but developing-economy national grids with division-level evening-peak reporting and co-visible limitation indicators are effectively absent from the collected corpus. Comparative studies that benchmark many deep architectures on common splits provide useful methodological guidance on model families [4], yet they seldom incorporate spatial graph structure or operational stress targets and therefore address only part of the planning problem faced by national-system operators.

A parallel literature stream addresses load shedding, under-frequency response, and emergency control rather than ex-ante day-ahead forecasting for planning. Approximately one quarter of the reviewed papers belong to this shedding and control cluster [10], [11]. Their objectives are optimising shedding sequences, preserving voltage or frequency stability, coordinating generator tripping, or managing islanded microgrids under disturbance. Datasets are often simulation-based, event-centric, or tied to specific subsystems and islanding scenarios. Methodologically, these studies employ particle-swarm and other metaheuristic optimisers, reinforcement learning, decentralised control rules, and predictive components that serve

real-time actuation rather than graded daily stress visibility. The distinction matters for task formulation: forecasting sparse or zero-inflated regional unmet-demand signals for planning is methodologically adjacent to shedding research but not equivalent to optimising control actions at disturbance timescales. Operational stress and reliability studies in the corpus address adjacent but distinct questions—transmission-level security assessment, semi-Markov reliability models, and vulnerability of demand-response schemes under cyber threat—but do not typically produce daily forecastable stress indices aligned with divisional demand series on national operational timelines.

Across the load-forecasting cluster, recurring limitations include weak external-validity reporting—particularly the absence of South Asian or Bangladesh division-level case studies—metadata-sparse conference abstracts that preclude reproducible dataset characterisation, and evaluation metrics that mix pooled and macro-aggregated definitions of coefficient of determination without explicit documentation. Unresolved challenges include adapting dense AMI-centric pipelines to daily national operational timelines with calendar gaps, modelling regional heterogeneity on footprints where a single division dominates the load budget, and coupling load forecasts with system-pressure indicators derived from limitation dynamics rather than treating shedding as a control output alone.

B. Graph-based Spatio-Temporal Forecasting

Graph neural networks and graph-attention architectures [7], [25] constitute a smaller but methodologically central subset of the reviewed corpus. Reported graph-based load models typically construct nodes from geographic areas, charging stations, integrated-energy-system zones, or residential feeders, and define edges through administrative proximity, infrastructure topology, or correlation in historical series. Spatio-temporal graph convolutional networks propagate information across fixed neighbourhoods at each timestep before or after temporal encoding; multi-graph and attention-based variants allow multiple edge types or learned weighting among neighbours. High-relevance comparators in the corpus combine graph structure with temporal convolution or attention for EV charging load, integrated multi-energy systems, ISO-level system operators, and residential transfer-learning settings [8], [9], [26].

Transformer-based spatio-temporal forecasting appears in a related subset of papers that either stack graph attention with temporal transformers or apply dual-channel spatio-temporal transformers to decoupled seasonal trends [15]–[17]. These methods assume that explicit relational structure improves forecasts when nodes exhibit measurable coupling, and they often evaluate on US or European residential benchmarks, wind-speed graphs, or charging-network topologies with qualitatively different spatial semantics from a nine-division national grid. Geographic adjacency is a common inductive bias: border-connected regions receive uniform or topology-derived edge weights. Data-driven correlation graphs appear less frequently as the sole spatial prior, and hybrid geographic–correlation constructions are seldom ablated

against correlation-only alternatives on developing-economy operational records.

Dataset assumptions in the graph-temporal cluster diverge markedly from the Bangladesh setting motivating the present work. Node counts range from small residential subgraphs to multi-energy system graphs; temporal resolution is often sub-hourly; and stress or shedding targets are rarely co-forecast with regional demand in the same graph-transformer pipeline. Shallow graph-convolution baselines without transformer capacity represent an important methodological floor in the broader literature [6], [21]: they test whether graph propagation alone suffices when temporal encoding is limited, and they highlight that graph conditioning is not automatically sufficient when the temporal branch cannot represent multi-day context on low-frequency daily series. Limitations recorded in the critical review include insufficient reporting of graph-construction splits—whether adjacency is estimated on training data only—sparse explainability of learned spatial weights, and limited evidence on when geographical priors help versus add noise on strongly correlated national demand series.

Unresolved challenges in graph-based spatio-temporal forecasting for power systems include selecting graph priors when inter-regional coupling is driven by shared national growth and seasonality rather than border adjacency alone; determining whether transformer capacity is uniformly necessary or conditional on graph topology; and extending graph-temporal models from residential or charging-network benchmarks to national division-level nodes with synchronised limitation covariates. None of the collected high-relevance graph or transformer studies address Bangladesh shedding context or daily OSI-style stress targets, leaving geographic and task-formulation gaps at the intersection of spatial coupling and operational constraint forecasting.

C. Multi-Task Learning

Multi-task learning in energy systems [14] occupies a narrow band of the reviewed literature relative to single-task load forecasting and shedding control. Reported formulations most often couple related energy vectors—electricity, heat, gas—or hierarchical multi-step horizons within a single sector, using shared encoders and task-specific heads with attention-based fusion. One hierarchical multi-energy study in the corpus applies multi-task structure to integrated energy system load forecasting with spatio-temporal attention [19], [26], illustrating that shared representations can exploit cross-vector structure when tasks are statistically related. However, explicit coupling of continuous multi-node regional demand with a daily operational stress index derived from shedding intensity, reserve margin, and limitation pressure is not represented among the collected comparators.

Task isolation remains the dominant pattern outside this small multi-task subset. Demand forecasting papers treat load as the sole supervised target; shedding papers optimise control or classify events; operational-stress and reliability studies focus on transmission assets, semi-Markov outage models,

or security assessment functionals rather than forecastable daily stress scalars for dispatch planning. Where auxiliary targets appear—such as controllability scores or composition forecasts—they are not paired with graph-level spatial encoders on national regional footprints. The reviewed opportunity catalogue records comparatively few explicit multi-task extension proposals, suggesting that joint demand-and-stress formulation is recognised as conceptually attractive but under-implemented in published graph-temporal energy models.

Methodological assumptions in multi-task energy learning include that shared trunks improve data efficiency when tasks share latent structure, and that loss weighting can be tuned to mitigate task interference. The literature provides limited guidance on balancing megawatt-scale robust regression against bounded stress targets in daily operational data, or on repairing stress-head collapse when naive multi-task weighting yields uninformative auxiliary outputs—issues that arise when stress labels are derived composites rather than dense shedding events. Sparse and zero-inflated regional load signals further complicate direct multi-target regression of shedding at the node level, motivating graph-level stress indices that aggregate system pressure without treating every division as a shedding classifier.

Limitations of the multi-task strand include scarce ablation of single-task versus multi-task ceilings on identical architectures, infrequent reporting of Pareto trade-offs when auxiliary tasks marginally degrade primary metrics, and absence of developing-economy national case studies with documented chronological splits. Unresolved challenges include determining when multi-task learning is preferable to separate pipelines for operators who require only demand or only stress visibility, calibrating task weights under class imbalance and composite-target semantics, and evaluating whether shared graph-temporal encoders transfer stress-relevant structure beyond what single-task models extract from global limitation covariates alone.

D. Explainable AI

Explainability for load and smart-grid forecasting is sparsely represented in the reviewed corpus [12]. Only a small fraction of collected papers report post-hoc or interpretable-by-design explanations; the majority of deep forecasting models provide no attribution analysis. Where explainability appears, it clusters into three methodological styles: SHAP or Shapley-value attributions applied to control or shedding policies—often in microgrid under-frequency settings with knowledge distillation [11], [18]; general interpretable-machine-learning frameworks for decentralised smart-grid prediction [13]; and review articles cataloguing feature-importance, SHAP, and partial-dependence practices for scalar load series without graph-temporal multi-task outputs [12].

Assumptions in the explainability literature often treat attribution as a supplement to control validation or consumer-facing interpretability rather than as operator-facing diagnostics for coupled regional demand and national stress forecasts. Graph-temporal models rarely export spatial attention,

coalition-grouped gradients, and dual-path comparison against engineered stress components in a single evaluation programme. Review-level work [12] acknowledges that explainability methods can disagree on feature rankings for time-series models, yet few empirical studies document that disagreement explicitly for multi-node graph transformers or report agreement between gradient attributions and permutation-based importance on megawatt demand targets.

Limitations include attributions evaluated on small validation samples without prospective operator studies, conflation of post-hoc sensitivity with physical causation, and explainability reported for single-task models while stress and demand explanations are not jointly audited against domain-decomposed indices. Review-level surveys note that method choice materially affects ranked drivers for load forecasts, reinforcing the need for multi-method reporting rather than single-map validation when models feed operator review. Unresolved challenges involve standardising multi-method attribution for dual-task dashboards, aligning model explanations with engineered stress components when composite targets mix shedding, reserve, and limitation channels, and extending microgrid-oriented SHAP workflows to national graph models with eleven feature coalitions and division-level node mass concentration.

E. Summary and Research Gap

Taken together, the reviewed literature establishes capable methods within each thematic silo—tabular and deep load forecasters on residential or industrial series, graph and transformer models on topology-defined networks, hierarchical multi-energy multi-task encoders, and isolated explainability studies—but seldom integrates them on the problem geometry addressed here. Four cross-cutting gaps emerge from the evidence-based gap matrix and critical review.

First, geographic and contextual validity is limited: no collected peer-reviewed spatio-temporal load or shedding forecasting study uses Bangladesh division-level smart-grid daily data [1], and developing-economy national grids remain under-represented relative to US and European AMI-centric benchmarks. Second, task formulation remains fragmented: continuous regional demand forecasting is rarely coupled with operational-stress-style targets in one multi-task graph-temporal model, while shedding research predominantly targets control and optimisation rather than day-ahead forecast-centric planning visibility. Third, methodological integration is incomplete: only a minority of papers combine graph-based spatial coupling with transformer-style temporal modelling for multi-node forecasting, and few co-design limitation-aware exogenous covariates—fuel, water, maintenance, temperature anomalies, and calendar structure—with spatio-temporal architectures or document chronological evaluation discipline with ablations and corrected inferential testing. Fourth, explainability is rarely embedded in coupled demand-and-stress graph forecasts with explicit multi-method reporting.

These gaps motivate the Correlation-Aware Multi-Task Forecasting Framework (PF-STGT) as a response aligned with

the identified voids rather than as an assertion of universal superiority over prior art. The framework targets joint next-day regional evening-peak demand and graph-level Operational Stress Index prediction on Bangladesh national operational records under leakage-safe chronological evaluation. It adopts a train-estimated correlation graph as an empirically testable spatial prior alternative to geography-first or hybrid constructions common in the literature, pairs graph-aware and temporal transformer encoding in a parallel-fusion architecture suited to multi-node daily windows, and applies an explicitly repaired multi-task loss formulation so that stress learning remains stable alongside megawatt demand regression. Post-hoc attribution is integrated as a reporting obligation—coalition-level gradients, permutation comparisons, attention export, and case-stratified dual-path stress review—rather than as an optional appendix, reflecting the corpus-wide shortage of explainability in graph-temporal energy forecasting.

The present study therefore occupies the intersection of four under-served areas in the reviewed literature: Bangladesh division-level smart-grid analytics, multi-task coupling of demand and operational stress, graph-temporal modelling with documented graph-prior ablation, and explainability analysis on frozen multi-task outputs. It does not claim to resolve every unresolved challenge identified above—external replication on other national grids, probabilistic uncertainty quantification, and prospective operator validation remain outside the scope of a single case study—but it provides a reproducible, evidence-locked foundation on which those questions can be examined. The methodology, experimental protocol, and empirical evaluation that instantiate PF-STGT under this positioning are presented in the following sections.

III. METHODOLOGY

A. Problem Definition and Task Formulation

Accurate short-horizon electricity demand forecasting is a core capability in modern smart power systems, where system operators must balance generation dispatch, reserve allocation, and network security under growing variability in load and supply conditions. The growing availability of historical operational power system data has motivated spatio-temporal learning frameworks that exploit coupling among regional loads and system-wide contextual signals. Beyond demand alone, operators also require visibility into operational stress arising from reserve shortfall, fuel and infrastructure limitations, and load-shedding pressure. The present study addresses both requirements for the Bangladesh national power system, where daily operational records support day-ahead forecasting over multiple regional divisions; dataset-specific properties are detailed in Section 6.2.

Consequently, the forecasting task is formulated as a supervised multi-task learning problem [14] over a set of spatial units indexed by graph nodes. Let $\mathcal{V} = \{1, \dots, N\}$ denote the node set, with N regional divisions. Time is indexed in daily steps t . At each forecast origin t , the model receives a lookback window of length T consecutive observed days and predicts targets at horizon H . Inputs and outputs are defined

over the window $[t - T + 1, t]$ for observations and at $t + H$ for supervision. The batch dimension is omitted where clear from context.

Inputs comprise two coupled tensors. Regional node features are denoted $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$, where F_n is the number of node-level features per region per day, encoding division-level historical demand signals, engineered lag and rolling statistics, and related regional observables. National and system-level context is represented by $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$, where F_g is the number of global features per day, including calendar indicators, grid aggregates, operational limitation variables, and national generation scalars. Together, $\mathbf{X}^{\text{node}}$ and $\mathbf{X}^{\text{global}}$ represent all information available at or before day t from which forecasts are produced.

Task 1 produces a regional demand vector $\hat{\mathbf{y}}^{\text{demand}} = (\hat{D}_1, \dots, \hat{D}_N)^\top \in \mathbb{R}^N$, with each $\hat{D}_r$ expressed in megawatts (MW). Task 2 produces a graph-level operational stress scalar $\hat{y}^{\text{osi}} \in [0, 1]$. The demand output assigns one next-day prediction per node; the stress output is a single system-wide score rather than a per-region vector.

Task 1 is regional demand forecasting. For each node $r \in \mathcal{V}$, the target is next-day regional evening-peak demand $D_r(t + H)$, measured in MW. Node-level targets are required because regional units exhibit distinct load scales and spatially coupled dynamics that a single national aggregate cannot represent. Regional supply was excluded as a Task 1 target because prior analysis documented strong demand–supply collinearity, rendering supply largely redundant for demand forecasting. Division-level load-shedding was not specified as a direct regression target because it forms a sparse, zero-inflated signal at the regional level; shedding intensity is incorporated instead through the composite stress target defined below.

Task 2 is prediction of the Operational Stress Index (OSI), a bounded system-health score that integrates shedding severity, generation reserve margin, and operational limitation pressure. OSI is constructed from three components normalised using train-split bounds only:

$$\begin{aligned} c_1 &= \frac{L_{\text{total}}}{D_{\text{total}}}, & c_2 &= 1 - \frac{GR}{\text{Highest_Generation}}, \\ c_3 &= \frac{\text{TOL}}{\text{Highest_Generation}}, \end{aligned} \quad (1)$$

$$\text{OSI} = \text{mean}(\text{minmax}_{\text{train}}(c_1), \text{minmax}_{\text{train}}(c_2), \text{minmax}_{\text{train}}(c_3)), \quad (2)$$

where L_{total} is total regional shedding (MW), D_{total} is total regional demand, GR is generation reserve, and TOL is total operational limitation. Continuous regression on OSI was adopted in preference to binary shedding labels or quantised stress classes because it preserves graded stress magnitude and represents reserve- and limitation-driven stress that may occur without a shedding event. The supervised target is $\text{OSI}(t + H)$.

Regional demand and OSI are learned jointly because the targets are operationally related yet statistically non-redundant.

Demand forecasts describe expected load levels across spatial units, whereas OSI summarises constraint-related system pressure through components c_1 – c_3 that are not fully encoded by demand alone. Joint estimation provides synchronised day-ahead visibility over load and stress from shared calendar and system context. A leakage policy applies to stress supervision: when predicting $\text{OSI}(t + H)$, same-day $\text{OSI}(t)$ is excluded from the input feature set; only information observable at or before t may be used.

Both tasks share a single-step forecast horizon $H = 1$ day. At origin t , the model predicts $\{D_r(t+1)\}_{r \in \mathcal{V}}$ and $\text{OSI}(t+1)$ from inputs observed through day t . A one-day horizon aligns with the daily reporting cadence of the operational timeline, matches the short-range temporal structure of regional demand series, and is consistent with the lag-based feature design employed in this study. Longer horizons are reserved for future multi-step extensions; aligning both tasks at $H = 1$ ensures that demand and stress predictions refer to the same future operating day.

The multi-task objective combines Task 1 and Task 2 losses. Let $\mathcal{L}_{\text{demand}}$ denote a robust regression loss on the MW scale—implemented as Huber loss [27] averaged over the N regional demand targets—and let $\mathcal{L}_{\text{osi}} = \ell_{\text{MSE}}(\hat{y}^{\text{osi}}, \text{OSI}(t + H))$ denote mean squared error on the bounded stress target. The joint training objective is

$$\mathcal{L} = \mathcal{L}_{\text{demand}} + \lambda_2 \mathcal{L}_{\text{osi}},$$

where $\lambda_2 > 0$ weights stress learning against demand fit. Task-weight calibration, demand-loss scaling, and optimisation details are specified in subsequent methodology subsections.

This subsection establishes a leakage-safe multi-task learning formulation whose objective is the joint prediction of regional electricity demand and the Operational Stress Index at a common forecast horizon. Under this formulation, both targets are estimated together from shared spatio-temporal inputs, subject to the leakage policy and combined loss defined above. The dataset, feature construction, graph representation, model architecture, and training protocol that instantiate this problem are described in the following subsections.

B. Dataset Description

The multi-task formulation in Section 6.1 is applied to historical operational records from the Bangladesh national power grid. The study draws on a publicly available Bangladesh Smart Grid dataset (Mendeley Data, Version 5) [1] comprising daily system and regional reporting variables collected over several years of grid operation. For modelling, a daily operational timeline derived from the published archive is adopted as the study’s primary operational timeline; key properties of this timeline are summarised in Table I.

Data source and granularity. The underlying archive provides one record per calendar day. Each record aggregates national generation and demand indicators together with division-level operational signals. The timeline spans 21 November 2019 to 30 December 2024, yielding 1,850 unique

daily observations over a 1,866-day calendar span. Sampling is predominantly at one-day intervals; a small number of calendar gaps are present in the series and are accounted for in later modelling stages. All 45 recorded fields are complete, with no missing entries in the audited source timeline. Table I lists the domain, regional coverage, and structural properties used in this study.

Spatial coverage. The spatial units in $\mathcal{V}$ correspond to $N = 9$ administrative divisions of Bangladesh: Barishal, Chattogram, Cumilla, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet. This partitioning matches the regional reporting structure reflected in the source data and supports node-level demand forecasting across geographically distinct load centres.

Recorded variables. The daily records combine calendar metadata, national generation and demand scalars, exogenous operational constraints, and regional power-balance signals. Calendar fields include the date, day of week, month, year, and holiday indicators. National-level quantities document evening-peak and day-peak generation, minimum and forecast generation, and demand measured at generation and substation endpoints. Operational constraint fields capture gas and liquid-fuel limitations, coal supply limitations, low water levels at Kaptai Lake, and capacity under plant shutdown or maintenance. A national temperature anomaly proxy for Dhaka is also recorded. At the regional level, each division is associated with three coupled signals: evening-peak demand, supply, and load (unmet demand / shedding) in megawatts. In total, the source timeline contains 42 numeric operational variables and three non-numeric calendar descriptors alongside the date index.

Target variables. Two supervised targets are derived from the recorded timeline in alignment with Section 6.1. For Task 1, the primary target is next-day regional evening-peak demand $D_r(t + H)$ for each node $r \in \mathcal{V}$, taken from the division-level demand fields. For Task 2, the Operational Stress Index is a graph-level score in $[0, 1]$ constructed from system-wide shedding intensity, generation reserve margin relative to highest available generation, and aggregate operational limitation pressure—the inputs to components c_1 – c_3 defined in Section 6.1. Regional supply and division-level shedding are retained in the source record because they inform system context and stress composition, but only demand and OSI serve as forecasting targets under the frozen task definition.

Dataset characteristics. Several properties make the timeline suitable for the joint demand–stress forecasting problem addressed here. First, the daily cadence matches the day-ahead horizon $H = 1$ and the operational reporting rhythm of the national grid. Second, the nine-division spatial structure provides a natural node set for regional load forecasting rather than a single national aggregate. Third, the coexistence of demand, shedding, reserve, and limitation variables within the same daily record enables construction of a composite operational stress indicator alongside load targets. Fourth, the series covers multiple years and full seasonal cycles, including holiday effects and periods of supply constraint, offering

variation in both nominal demand levels and stress regimes.

Prior data review documented strong coupling between regional demand and supply series, motivating demand-only targets for Task 1, and a sparse, zero-inflated distribution of regional load variables, motivating incorporation of shedding through OSI rather than as a direct per-node regression target. The timeline further exhibits inter-regional correlation in demand dynamics, supporting graph-based spatio-temporal modelling in subsequent subsections.

Suitability for this study. The dataset is appropriate for this study because it provides synchronised national- and regional-level operational variables within each daily record, supporting joint demand and operational stress forecasting under the formulation in Section 6.1. Collectively, the source dataset and derived timeline provide the regional coverage, temporal extent, variable richness, and dual-target structure required by that formulation. The dataset offers a documented case study of a developing-economy national grid in which day-ahead regional demand and operational stress must be assessed jointly from historical operational power system data. Feature construction, graph representation, and model specification built on this timeline are described in the following methodology subsections.

C. Data Preprocessing

The operational timeline described in Section 6.2 is prepared for downstream modelling through a two-stage preprocessing pipeline that precedes feature engineering. The first stage performs conservative scientific cleaning to validate and correct the source record while preserving genuine operational events. The second stage converts the cleaned timeline into a model-ready representation through chronological ordering, categorical encoding, and numeric standardisation, with all learnable transformations restricted to training data to prevent temporal leakage. No lag features, rolling statistics, graph structures, or supervised targets are created in this stage.

Data validation and integrity assessment. Preprocessing begins with a read-only audit of the published raw archive to establish record counts, field types, and temporal structure without altering source values. The audit confirmed 1,850 daily rows and 45 fields, with 42 numeric operational measurements and three non-numeric calendar descriptors indexed by date. Missing-value assessment across all fields returned zero absent cells. Duplicate inspection found no repeated rows and no duplicate dates. Negative measurements and implausible temperature values were absent. These checks establish that the archive is complete at the cell level and suitable for time-ordered forecasting, while deferring any modification of the raw file to the cleaning stage.

Scientific data cleaning. The cleaning stage ingests the published archive and produces a cleaned intermediate dataset of dimensions $1,850 \times 45$. Cleaning follows a conservative rule: only impossible, corrupted, duplicated, or invalid records are corrected or removed; statistically rare but operationally genuine events are retained. No rows were removed. The date field was validated and converted to a standard temporal

representation without changing calendar values. Timestamp verification confirmed that all dates are parseable, unique, and strictly monotonic increasing over the range 2019-11-21 to 2024-12-30. Chronological consistency checks verified agreement between the date index and the recorded year and month fields in every row. Six inconsistent day-of-week labels were corrected deterministically from the authoritative date index; no numeric measurements were altered.

Calendar continuity was assessed against a daily index. Seventeen missing calendar days were documented in the series; these gaps were not imputed or synthetically filled during cleaning, so only observed daily records are retained. This policy preserves the authentic reporting cadence and avoids introducing artificial observations that would distort lag-based forecasting. Physical-consistency review identified regional accounting mismatches and generation-ordering discrepancies in a subset of rows; these patterns were classified as reporting or metering differences rather than impossible values and were preserved as informative operational signals. Row and column counts were unchanged between input and output, confirming that cleaning altered only invalid metadata rather than measurement content.

Model-ready preprocessing. The cleaned intermediate timeline serves as the sole input to the model-preparation stage. Records are sorted in ascending chronological order to enforce temporal ordering required for causal forecasting. Three categorical fields—day of week, holiday name, and holiday category—are one-hot encoded using categories observed in the training partition only, with unseen validation or test categories assigned zero contribution. This encoding supplies a fixed numeric representation of calendar and holiday context without manual category collapsing.

Forty-one numeric operational fields, comprising national generation and demand scalars, limitation indicators, temperature, calendar ordinals (year and month), and all regional demand, supply, and load measurements, are standardised by z-score scaling with parameters learned exclusively from the training partition. Validation and test partitions are transformed using the frozen training statistics only. Standardisation places heterogeneous MW-scale quantities on a common scale for subsequent learning while preventing held-out data from influencing the fitted parameters. The date index is retained as an unscaled temporal key. Only existing source fields are transformed; no new predictive attributes are introduced at this stage.

Leakage prevention. Temporal leakage is controlled at two levels during preprocessing. First, chronological ordering is preserved throughout cleaning and transformation so that no future observation precedes a past one in the stored timeline. Second, all learnable preprocessing parameters—the one-hot category sets and scaling means and standard deviations—are estimated on the training partition only; validation and test data receive transformation with frozen training parameters. Cleaning and model-preparation stages remain separable, so encoding and scaling cannot retroactively alter the cleaned record. Same-day stress indices and other supervision-derived

quantities are not injected during this stage; the leakage policy for OSI inputs defined in Section 6.1 is enforced in subsequent feature construction.

Preprocessing outputs. The pipeline delivers three products for later methodology stages: a validated cleaned intermediate dataset that preserves authentic operational events; chronologically partitioned processed records in encoded and standardised form; and a reproducible preprocessing specification together with a serialised transformation pipeline that freezes encoding and scaling rules. Together, these outputs provide a leakage-aware foundation on which lag-based node features, global context variables, graph construction, and multi-task training are built in the following subsections.

D. Feature Engineering

The leakage-aware preprocessing outputs described in Section 6.3 are extended into a structured feature set that supplies the spatio-temporal inputs defined in Section 6.1. Feature engineering transforms the cleaned operational timeline and its encoded baseline representation into derived predictors that capture autoregressive load dynamics, calendar structure, grid-wide stress context, and regional heterogeneity. The design prioritises causal observability: every engineered quantity at day t is computable from information available at or before t , with no reliance on future rows or held-out partition statistics. In total, 65 engineered features are constructed and merged with the preprocessed baseline by date. Raw source measurements and their preprocessed encodings are distinguished throughout from quantities constructed by deterministic transformation rules.

Design objectives. The feature programme serves three methodological goals aligned with the joint demand–stress formulation. First, it encodes short-horizon temporal structure appropriate to a one-day forecast horizon and a seven-day input window $T = 7$, including autoregressive demand signals and gap-aware offsets that respect irregular calendar spacing. Second, it represents both node-local load state and national system context within each daily record, enabling the model to relate regional forecasts to grid-wide limitation and reserve conditions. Third, it enforces a strict leakage policy: same-day Operational Stress Index values are excluded from the model input specification even though an OSI-derived feature is computed for target construction and diagnostic use, consistent with Section 6.1.

Raw variables versus engineered features. Raw variables comprise the division-level evening-peak demand, supply, and load measurements, national generation and demand scalars, operational limitation indicators, temperature, and calendar descriptors described in Section 6.2. Preprocessing supplies encoded and standardised forms of these fields without creating new attributes. Engineered features are deterministic functions of raw and preprocessed quantities—lags, rolling statistics, normalised shares, grid aggregates, cyclical calendar encodings, and composite stress indicators—that are computed on the chronological timeline and merged with the baseline representation by date. Continuous engineered quantities are

standardised using training-partition statistics only; a binary shedding-presence indicator is retained without scaling.

Temporal feature engineering. Four global temporal descriptors encode seasonality, long-run drift, and irregular sampling. Day-of-year sine and cosine features provide smooth annual periodicity without discontinuities at year boundaries. A monotonic trend index captures slow structural change across the multi-year record. An inter-observation gap feature records the elapsed days between consecutive observed records, explicitly signalling calendar discontinuities rather than treating missing days as zero-load intervals. At each node, three lag families supply autoregressive structure: one-day and seven-observation demand lags, and one-day load lags. Lags use observed-row offsets on the sorted timeline rather than fixed calendar offsets, so a seven-observation lag advances seven recorded days even when calendar gaps intervene. Seven-day demand rolling means are computed over the seven preceding observed rows with the current day excluded, yielding a past-only smoothed load level that complements point lags for short-horizon forecasting.

Node-level engineered features. The node input specification comprises three conceptual groups. A contemporaneous power-balance group carries processed demand, supply, and load at each division, anchoring the node to its current operational state. An autoregressive temporal group supplies demand lags at one and seven observations, a one-observation load lag, and a seven-observation demand rolling mean, encoding short-term persistence and weekly structure aligned with the input window. A regional derived group expresses each division’s relative role in the national system: demand share $D_r(t)/\sum_j D_j(t)$, which normalises heterogeneous scales to a common simplex representation, and load intensity $L_r(t)/\max(D_r(t), \varepsilon)$ with $\varepsilon = 1$ MW, which expresses unmet demand relative to local load without treating sparse shedding zeros as informative magnitude on their own. These three groups yield $F_n = 9$ channels per region in the node tensor $\mathbf{X}^{\text{node}}$.

Global engineered features. The global input specification comprises six conceptual groups. A temporal and calendar group includes the four engineered descriptors above—cyclical day-of-year encodings, trend index, and inter-observation gap—together with a holiday-category summary derived by consolidating the four one-hot holiday-category channels produced in Section 6.3 into a single ordinal indicator for model input. A grid-aggregate group sums total regional demand and total regional load and computes generation reserve as the difference between highest available generation and evening-peak demand at the generation endpoint, summarising system-wide balance and headroom. An operational group aggregates gas, coal, water, and maintenance limitation indicators into total operational limitation, records whether any division reports positive shedding, and—during feature construction only—forms a composite stress score from shedding intensity, reserve margin, and limitation pressure using train-split normalisation bounds; this composite is excluded from model inputs to prevent same-day OSI leakage when forecasting $\text{OSI}(t + 1)$.

A weather group expresses temperature anomaly relative to train-estimated monthly means. A limitation group retains the four preprocessed operational-constraint scalars from the baseline representation. A national generation group retains two preprocessed generation-endpoint scalars. Together, these groups yield $F_g = 17$ channels in the global tensor $\mathbf{X}^{\text{global}}$ at each timestep.

Operational feature representation. Operational engineering targets the constraint drivers underlying Task 2 without substituting the supervised OSI label into the input set. Total operational limitation condenses fuel, infrastructure, and maintenance constraints into a single MW-scale pressure index. Generation reserve links available capacity to contemporaneous demand. The binary any-division shedding indicator flags system-wide unmet-demand events that may precede graded stress escalation. The engineered stress composite mirrors the component structure of Section 6.1—shedding ratio, reserve shortfall, and limitation pressure—but is withheld from $\mathbf{X}^{\text{global}}$ at inference time so that stress predictions at $t + 1$ cannot access a normalised stress score already computed at t .

Feature grouping. The implemented channels are organised into eleven coalitions that separate raw regional blocks from engineered groups: regional demand, supply, and load blocks (G1–G3); engineered lags and rolling statistics (G4); regional share and intensity features (G5); calendar and trend descriptors (G6); grid aggregates (G7); the limitation stack (G8); weather anomaly (G9); national generation scalars (G10); and the shedding indicator (G11). This grouping maps each coalition to either node or global scope while preserving the distinction between contemporaneous measurements and derived predictors. The coalition definitions support grouped attribution analysis in Section 8.5 (Table VII).

Leakage-aware feature design. All lag and rolling computations apply strictly backward shifts on the chronological timeline. Monthly temperature means, OSI normalisation bounds for the composite stress feature, and standardisation parameters for continuous engineered fields are estimated on the training partition only. The Operational Stress Index at day t is never included among the seventeen global model inputs. Demand and load histories through day t remain admissible for forecasting at $t + 1$. Calendar gaps are propagated through the gap-day feature and gap-aware lags rather than imputed.

Final feature representation. At each forecast origin, the model receives a lookback window of $T = 7$ days. Node inputs form $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ with $N = 9$ and $F_n = 9$, assembling the nine node channels for every division at each day in the window. Global inputs form $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$ with $F_g = 17$, broadcasting national and system context across the same window. Raw and preprocessed measurements enter through the contemporaneous demand–supply–load triplets, limitation scalars, generation scalars, and encoded calendar fields; engineered quantities supply autoregressive, seasonal, and aggregate context. This representation instantiates the input tensors defined in Section 6.1 and provides the feature foundation on which graph construction and model specifica-

tion are described in the following subsections.

E. Graph Construction

The node and global feature tensors defined in Section 6.4 encode division-level and system-wide signals along the temporal axis but do not, by themselves, specify how spatial coupling among regional loads should be represented. Regional evening-peak demand series in the Bangladesh dataset exhibit strong positive inter-division correlation driven by shared national growth, seasonality, and common calendar shocks, as documented during exploratory analysis. Treating each division as an independent univariate time series would ignore this coupling and forego an explicit mechanism for propagating load information across the national footprint; separate models per division would also duplicate global context already encoded in $\mathbf{X}^{\text{global}}$ without enabling cross-division information flow at the regional scale. A graph representation is therefore adopted: each administrative division is a vertex, and weighted edges encode the strength of historical demand co-movement between division pairs. Graph construction converts the validated feature timeline into a fixed spatial prior that accompanies the spatio-temporal inputs at every forecast origin.

Node definition. The vertex set $\mathcal{V}$ contains $N = 9$ nodes, one per administrative division—Barishal, Chattogram, Cumilla, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet—in fixed alphabetical order for reproducible indexing. Each node is aligned with the corresponding row of the node feature tensor $\mathbf{X}^{\text{node}}$ described in Section 6.4: contemporaneous demand, supply, and load; autoregressive lags and rolling means; and regional share and intensity descriptors. National and system-level quantities remain global context in $\mathbf{X}^{\text{global}}$ rather than separate graph vertices, preserving a clear separation between node-local load state and grid-wide conditioning.

Spatial dependency modelling. Spatial dependence is modelled through undirected edges whose weights reflect pairwise statistical association in regional demand. During graph design, three construction strategies were specified—geographical adjacency, correlation-thresholded adjacency, and a hybrid of both—so that domain priors and data-driven coupling could be compared under a common node set. For the proposed framework, the correlation-only strategy is adopted as the definitive edge rule; alternative formulations are described in a later subsection, and the correlation-graph construction steps are detailed below.

Correlation computation. Pairwise edge weights are derived from Pearson correlation coefficients ρ_{ij} computed on regional evening-peak demand series using training-partition observations only ($n = 1,295$ daily records). Restricting correlation estimation to the training partition prevents held-out periods from influencing the spatial prior. Demand was selected as the correlation basis because it is the primary Task 1 target and exhibits the strongest and most stable inter-regional association in the exploratory record. Train-period demand correlations are uniformly positive across division

pairs, with the weakest associations confined to a small subset of pairs; most divisions co-vary strongly over shared seasonal and growth cycles rather than evolving independently.

Edge definition and sparsification. An undirected edge (i, j) is retained when $\rho_{ij} \geq \tau$, with correlation threshold $\tau = 0.65$. Edge weight equals ρ_{ij} for retained pairs and zero otherwise. The threshold follows the correlation cut-off adopted during exploratory analysis and graph design: it retains pairs with substantively positive demand co-movement, excludes the weakest associations, and yields a connected graph suitable for spatial message passing. Under this rule, 33 of 36 possible undirected pairs are retained (91.7% pair density). Self-loops are excluded by setting diagonal entries to zero, so each node does not receive a trivial self-message through the spatial prior. The resulting topology is symmetric, non-negative, and connected.

Graph normalisation. Raw correlation weights are row-normalised to produce the adjacency matrix $\mathbf{A}$ used in the proposed framework. Each row is divided by its sum so that outgoing edge weights from a node form a convex combination summing to unity. From a methodological standpoint, row normalisation expresses each division’s spatial neighbourhood as a weighted mixture of its neighbours rather than an unscaled sum of correlation magnitudes, which stabilises cross-division aggregation when node degree and edge strength vary. It also prevents high-degree nodes from dominating neighbour contributions solely because they retain more edges. The normalised matrix is fixed after training-partition estimation and applied unchanged across validation and test periods.

Static versus dynamic graph. The adjacency matrix is static: edge structure and weights are estimated once from the full training timeline and held fixed thereafter. Time-varying graph refresh—for example, through rolling correlation windows—was considered during graph design but deferred to keep the spatial prior reproducible and leakage-safe. Dynamic national context instead enters through the global feature channels and the temporal dimension of $\mathbf{X}^{\text{node}}$, while $\mathbf{A}$ supplies a stable map of historical inter-division demand coupling.

Alternative graph formulations. The geographical graph connects border-adjacent divisions with uniform binary edges (21 undirected pairs; 58.3% density), row-normalised for aggregation. This variant encodes an administrative-border prior on plausible power-exchange neighbourhoods but assigns equal weight to all geographic links regardless of observed coupling strength. The hybrid graph retains an edge when either a geographical link exists or $\rho_{ij} \geq 0.85$, with weight ρ_{ij} (24 undirected pairs; 66.7% density). This variant augments the geographic skeleton with the strongest non-border demand associations. Neither alternative enters the proposed framework, which employs the correlation-only rule with $\tau = 0.65$ described above.

Final graph representation. Graph construction yields a row-normalised adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ together with an attention-bias matrix obtained by a deterministic transformation of $\mathbf{A}$ —log-scaling positive entries and masking

absent edges—for use by the subsequent transformer-based architecture. The bias matrix is not an independent graph specification; it re-expresses the same normalised adjacency structure in the form required for additive spatial masking in the model. Both matrices are fixed, train-estimated artefacts aligned to the alphabetical node order. At each forecast origin, $\mathbf{A}$ accompanies the windowed node and global feature tensors, supplying the spatial coupling structure on which the spatio-temporal forecasting model described in the following subsections operates. By encoding inter-division demand association explicitly, the graph representation enables coordinated regional forecasts that respect observed national load synchronisation rather than treating each division as an isolated series.

F. Proposed STGT Architecture

The graph prior defined in Section 6.5 supplies spatial structure; the feature tensors defined in Sections 6.1 and 6.4 supply temporal and nodal observations. The proposed architecture—the Correlation-Aware Multi-Task Forecasting Framework, implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT)—integrates these representations within a single neural model that jointly forecasts next-day regional demand and the Operational Stress Index. Architectural philosophy follows a parallel dual-branch design: one branch emphasises inter-division coupling at each timestep through graph-aware attention, while a second branch emphasises within-division temporal dynamics across the input window. The branches operate on a common embedded representation and are combined through learnable gating before task-specific prediction heads. This organisation separates data encoding, graph conditioning, spatio-temporal encoding, and multi-task decoding into distinct stages, enabling each module to address a specific modelling requirement of the Bangladesh joint forecasting problem. The overall framework is illustrated in Fig. 1.

Model inputs and representational layers. Three input layers feed the architecture, and each serves a distinct role. The data representation comprises the node tensor $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ and the global tensor $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$, constructed as described in Section 6.4 for a lookback window of $T = 7$ days ending at forecast origin t . The graph representation comprises the row-normalised adjacency matrix $\mathbf{A}$ and the derived attention-bias matrix from Section 6.5; these artefacts are fixed across forward passes and do not vary with the input window. The neural architecture maps embedded inputs through spatial and temporal encoders, fuses the resulting hidden states, and decodes task outputs. Supervision targets—regional demand at $t+1$ and $\text{OSI}(t+1)$ —are not model inputs; the leakage policy in Section 6.1 excludes same-day OSI from the feature set when forecasting next-day stress.

Input embedding. Raw feature tensors are first projected into a shared latent space of dimension d_{model} . Node channels are linearly mapped and augmented with a learnable regional identity embedding so that the model can distinguish divisions that share similar load scales. Global channels are linearly

mapped and broadcast across nodes at each timestep, injecting national calendar, limitation, and generation context into every spatial unit. A learnable positional encoding over the T window timesteps is added so that the model can discriminate recent from older observations within the lookback. The resulting initial hidden state $\mathbf{H}_0 \in \mathbb{R}^{T \times N \times d_{\text{model}}}$ unifies node-local, global, and temporal-index information in a single tensor that both encoder branches consume. Embedding is required because heterogeneous engineered features and graph topology operate in different mathematical spaces; projection places all signals in a common representation before attention-based mixing.

Graph integration. The correlation graph enters the spatial encoder through an additive attention bias derived from $\mathbf{A}$. For each ordered node pair (i, j) , positive adjacency weights contribute a log-scaled bias term to the attention score, while absent edges receive a large negative mask so that message passing is restricted to the correlation-supported neighbourhood. This mechanism couples learned attention patterns with the train-estimated spatial prior from Section 6.5 rather than relying on free-form pairwise weights alone. Graph integration is applied independently at each of the T timesteps in the spatial branch, allowing inter-division information exchange to vary day by day within the window while the underlying adjacency structure remains static.

Spatial encoder. The spatial branch is a graph transformer [7], [17] that applies multi-head self-attention over the N nodes at each timestep. Given hidden states at timestep t , attention scores combine scaled query-key inner products with the graph bias matrix, and the softmax-normalised weights aggregate value vectors across nodes:

$$\text{Attn}_{\text{spatial}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{adj}}\right) \mathbf{V},$$

where $\mathbf{B}_{\text{adj}}$ denotes the attention-bias matrix derived from $\mathbf{A}$. Stacked encoder layers with residual connections and position-wise feed-forward transformations refine $\mathbf{H}_0$ into a spatially encoded sequence $\mathbf{H}_{\text{spatial}} \in \mathbb{R}^{T \times N \times d_{\text{model}}}$. A graph transformer was selected over fixed-weight graph convolutions because division coupling strength varies across pairs and because attention weights provide an interpretable record of spatial information flow. The spatial encoder addresses the limitation of treating regions as independent: it propagates demand-relevant signals along correlation-weighted edges at each day in the lookback window.

Temporal encoder. The temporal branch is a transformer encoder [17] applied independently to each node’s T -step sequence, with parameters shared across nodes for efficiency. At each node, multi-head self-attention operates over timesteps, capturing short-range autoregressive structure and weekly periodicity aligned with the seven-day input window and lag-based features in Section 6.4. The encoder outputs $\mathbf{H}_{\text{temporal}} \in \mathbb{R}^{T \times N \times d_{\text{model}}}$ with the same shape as the spatial branch. A full transformer encoder was preferred over a single shallow attention layer because the operational record exhibits multi-

day seasonal structure within the window. Causal masking is not applied: the window contains only observations at or before the forecast origin, so ex-post encoding of the full lookback is consistent with the day-ahead forecasting setting.

Feature fusion. Spatial and temporal branches encode complementary aspects of the input—cross-division coupling and within-division dynamics—and are integrated through parallel fusion rather than a strict sequential pipeline. A learnable gate computed from the concatenated branch outputs forms a convex combination at each timestep and node:

$$\mathbf{H}_{\text{fused}} = \mathbf{G} \odot \mathbf{H}_{\text{spatial}} + (1 - \mathbf{G}) \odot \mathbf{H}_{\text{temporal}},$$

where $\mathbf{G} = \sigma(\text{Linear}([\mathbf{H}_{\text{spatial}}; \mathbf{H}_{\text{temporal}}]))$ and $\odot$ denotes element-wise multiplication. Parallel fusion allows the model to emphasise spatial or temporal evidence depending on context—for example, synchronised national load movement versus division-specific short-term persistence—without forcing one inductive bias to precede the other. Sequential spatial-then-temporal or temporal-then-spatial designs were rejected during architecture design because they delay one form of coupling and can blur the temporal structure before cross-division mixing, or defer spatial propagation of same-day regional shocks.

Shared latent representation. The fused sequence is reduced to a per-node shared embedding by selecting the last observed timestep in the window:

$$\mathbf{H}_{\text{shared}} = \mathbf{H}_{\text{fused}}[:, T, :, :] \in \mathbb{R}^{N \times d_{\text{model}}}.$$

This tensor constitutes the common latent state from which both tasks are decoded. A graph-level readout vector is additionally formed by mean-pooling $\mathbf{H}_{\text{shared}}$ across nodes and concatenating the result with the embedded global context at the final timestep, yielding a compact system-wide summary for stress prediction. The shared representation enforces multi-task information sharing: demand and OSI heads draw on the same spatio-temporal encoding of the input window rather than independent models that would duplicate feature extraction.

Demand prediction head. Task 1 is decoded through a dedicated per-node linear head that maps each row of $\mathbf{H}_{\text{shared}}$ to a scalar next-day demand forecast in megawatts, producing the regional output vector $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ defined in Section 6.1. A node-specific linear mapping is sufficient because $\mathbf{H}_{\text{shared}}$ already encodes division identity through the regional embedding and spatial attention; the head’s role is to project the shared latent state onto the MW scale for each region. This design yields nine heterogeneous forecasts from one trunk while preserving parameter efficiency relative to nine independent sequence models.

OSI prediction head. Task 2 is decoded through a graph-level head that consumes three complementary views of the shared state: the mean-pooled graph summary concatenated with the final-timestep global embedding, and the flattened node-level shared representation across all divisions. A feed-forward network with a bounded output activation maps this

combined vector to a single scalar $\hat{y}^{\text{osi}} \in [0, 1]$. The OSI head requires both aggregated system context and division-resolved latent states because operational stress in this formulation is a national score influenced by spatially distributed load and constraint patterns. Restricting the stress decoder to graph-pooled information alone would discard node-level variation retained in $\mathbf{H}_{\text{shared}}$; restricting it to a single node would fail to represent system-wide stress.

Multi-task learning mechanism. Demand and OSI prediction share the input embedding, both encoder branches, the fusion module, and the $\mathbf{H}_{\text{shared}}$ tensor; only the final decoding layers are task-specific. Shared encoding is warranted because both targets depend on the same calendar, limitation, and load history within the lookback window, yet the targets are not redundant: demand is node-resolved and continuous on the MW scale, whereas OSI is graph-resolved and bounded. The dual-head structure implements the multi-task formulation of Section 6.1 without duplicating the spatio-temporal trunk. Task interaction and loss balancing are specified separately; the architecture itself supplies the shared representational substrate on which joint learning operates.

Architectural rationale. The proposed PF-STGT design addresses three requirements identified during methodology development. First, correlation-only graph conditioning—rather than treating divisions as independent series—supplies a data-driven spatial prior matched to the strong inter-regional demand coupling documented in Sections 6.2 and 6.5. Second, parallel spatial and temporal transformer branches jointly model cross-division and within-division dynamics that co-exist in daily grid operations. Third, a shared latent core with specialised demand and stress heads implements joint day-ahead forecasting of load and operational stress from one spatio-temporal pass over the input window. Together, these choices define the Correlation-Aware Multi-Task Forecasting Framework that instantiates the problem formulation, feature representation, and graph prior developed in the preceding subsections; training protocol and optimisation details are described in subsequent methodology subsections.

G. Training Strategy

The PF-STGT architecture in Section 6.6 is trained on chronologically partitioned samples that respect the leakage policies established in Sections 6.1 and 6.3. Training philosophy prioritises temporal integrity: future observations never influence past inputs, preprocessing and graph parameters are frozen from the training partition, and model selection uses only the validation period while the test period remains unseen until final evaluation. Samples are constructed as sliding windows over each split independently, with multi-task targets aligned to a common forecast horizon. This protocol converts the engineered feature timeline and static graph prior into supervised training examples without crossing partition boundaries or contaminating held-out periods.

Chronological partition. The operational timeline is divided into training, validation, and test subsets by ascending date, following the chronological split applied during prepro-

cessing. The training partition spans 21 November 2019 to 15 June 2023 (1,295 daily records); validation spans 16 June 2023 to 19 March 2024 (277 records); and test spans 20 March 2024 to 30 December 2024 (278 records). Partition boundaries are strictly non-overlapping: the latest training date precedes the earliest validation date, and the latest validation date precedes the earliest test date. Chronological partitioning is required because random shuffling of days would place future grid states in the training set and inflate performance estimates. The split ratios and date ranges are summarised in Table I.

Sequence generation and warm-up exclusion. Training examples are generated independently within each partition. For a partition containing R chronologically ordered daily records, a valid forecast origin must satisfy two constraints. First, the lookback window of length $T = 7$ must lie entirely within the partition and contain only rows with complete engineered features. Second, the supervision index must lie one horizon step ($H = 1$) ahead of the window end and remain inside the same partition. The first seven rows of each partition are excluded as warm-up because seven-observation demand lags and seven-day rolling means are undefined at the series start; this policy aligns with the lag-based feature design in Section 6.4. The resulting numbers of valid sliding-window samples per partition are reported in Table I. Calendar gaps in the source timeline are preserved: windows span consecutive observed rows within a partition rather than imputed calendar days.

Sliding-window construction. Each training example is built by sliding a fixed-length window one observed row at a time. For forecast origin index t_{end} , the input window covers days $[t_{\text{end}} - T + 1, t_{\text{end}}]$ and supplies the node tensor $\mathbf{X}^{\text{node}}$ and global tensor $\mathbf{X}^{\text{global}}$ described in Section 6.4. The static adjacency matrix $\mathbf{A}$ and its derived attention-bias matrix from Section 6.5 are attached to every sample because the graph prior does not vary across origins. Window construction is identical across partitions; only the allowable range of t_{end} changes with partition size and boundary position. Sliding-window enumeration maximises the number of supervised examples available within each period without altering the one-day forecast horizon.

Multi-task target preparation. Supervision is prepared separately from model inputs. For Task 1, regional evening-peak demand at day $t_{\text{end}} + H$ is extracted from the cleaned interim timeline in raw megawatts for each of the nine divisions, producing the target vector $\mathbf{y}^{\text{demand}} \in \mathbb{R}^N$. For Task 2, the Operational Stress Index at the same future day is computed from the component formulae in Section 6.1 using train-partition min-max bounds for normalisation; the resulting scalar $\text{OSI}(t_{\text{end}} + H)$ lies in $[0, 1]$. Demand targets use unscaled MW values from the clean operational record so that the demand head learns on a physically interpretable scale. OSI component bounds are fit once on the training partition and frozen, consistent with the leakage policy that forbids using validation or test statistics in target construction. Same-day OSI at t_{end} is never included among input features when predicting $\text{OSI}(t_{\text{end}} + H)$.

Batch formation. Training iterates over mini-batches of windowed samples drawn from the training partition. Within a batch, node and global feature tensors are stacked along the batch dimension; demand and OSI targets are stacked correspondingly. The adjacency and attention-bias matrices are shared across all samples in a batch because the graph is static. Training batches are shuffled each epoch to vary the order of independent window samples; this shuffling does not alter the chronological ordering of observations within any individual window. Validation and test loaders preserve chronological sample order so that temporal structure remains inspectable. Batch construction therefore separates variable per-origin features from fixed graph artefacts shared across the national grid.

Validation protocol. During training, model parameters are updated using only the training partition. After each epoch, performance is assessed on the validation partition using the same windowing and target rules, without parameter updates. Validation metrics include macro demand mean absolute error over the nine regions and stress mean absolute error for the graph-level OSI target. For the multi-task PF-STGT, checkpoint selection is guided by a balanced composite validation score that reflects both demand and stress prediction error, so that model selection does not optimise a single task at the expense of the other. The test partition is not used during training, hyperparameter selection, or early stopping; it is reserved for the held-out evaluation reported in Section 8.

Early stopping and checkpoint selection. Training proceeds epoch by epoch until a convergence criterion is met. Early stopping monitors the composite validation score with a patience-based rule: if the score fails to improve by a minimum margin for a fixed number of consecutive epochs, training terminates. When validation performance improves, the best-performing model checkpoint is retained based on validation performance. Checkpoint selection is therefore tied exclusively to validation error, not training loss. The checkpoint corresponding to the lowest composite validation score is used as the candidate model for test evaluation.

Reproducibility considerations. Training reproducibility is supported at three levels. First, a fixed random seed governs weight initialisation, batch shuffling, and stochastic operations so that repeated runs with the same configuration are comparable. Second, the frozen datasets, engineered feature representations, and graph specifications developed in preceding methodology stages are verified for consistency before training begins, ensuring that each run consumes the same inputs. Third, partition boundaries, window length, horizon, warm-up exclusion, and sample-index generation are defined in shared constants and applied identically across training, validation, and test loaders. Together, these controls bind the training strategy to the data and graph representations established earlier in the methodology. Table II summarises the frozen optimisation hyperparameters and multi-task loss specification.

H. Loss Function and Optimization

The training protocol in Section 6.7 defines how supervised windows are formed, partitioned, and validated; this subsection specifies the objective function and numerical optimisation procedure used to fit the PF-STGT parameters. Optimisation is formulated as minimisation of a weighted multi-task loss over mini-batches drawn from the training partition. Demand supervision remains on the physically interpretable megawatt scale at evaluation time, while the training objective applies a scale-aware transformation so that gradient updates to the demand and stress heads remain numerically commensurate. Parameter updates follow a fixed recipe—Adam with weight decay, adaptive learning-rate reduction, and gradient-norm clipping—chosen for stability on the modest daily sample size and dual-output architecture described in Sections 6.1 and 6.6.

Multi-task objective. Let $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ denote the vector of predicted regional evening-peak demands and $\hat{y}^{\text{osi}} \in [0, 1]$ the predicted Operational Stress Index for a single forecast origin, with corresponding targets $\mathbf{y}^{\text{demand}}$ (MW) and y^{osi} prepared as in Section 6.7. The joint training objective extends the formulation introduced in Section 6.1 by separating raw demand fit from stress fit and introducing an explicit demand-scale normalisation before task weighting:

$$\mathcal{L} = \frac{\mathcal{L}_{\text{demand}}^{\text{raw}}}{100} + \lambda_2 \mathcal{L}_{\text{osi}},$$

where $\lambda_2 = 20$ is the frozen stress-task weight and the divisor 100 (MW) maps the demand loss term to a magnitude comparable with the bounded OSI loss. Minimisation is performed over all trainable parameters θ of the PF-STGT model by mini-batch optimisation using the Adam optimiser [28] with batch size $B = 32$.

Demand loss. Regional demand errors are aggregated with a macro-averaged Huber loss over the $N = 9$ divisions:

$$\mathcal{L}_{\text{demand}}^{\text{raw}} = \frac{1}{N} \sum_{r=1}^N \text{Huber}_{\delta}(\hat{y}_r^{\text{demand}}, y_r^{\text{demand}}),$$

where $\text{Huber}_{\delta}(a, b)$ is the element-wise Huber function with threshold $\delta = 1.0$ MW:

$$\text{Huber}_{\delta}(a, b) = \begin{cases} \frac{1}{2}(a - b)^2, & |a - b| \leq \delta, \\ \delta(|a - b| - \frac{\delta}{2}), & \text{otherwise.} \end{cases}$$

Huber regression was selected in preference to pure squared error because evening-peak demand exhibits occasional upper-tail excursions; the quadratic region penalises moderate errors smoothly, while the linear region limits the influence of large outliers that would otherwise dominate batch gradients. Macro-averaging across regions assigns equal weight to each division in the loss, consistent with the node-level forecasting objective in Section 6.1. Targets remain in raw megawatts during training so that the demand head learns on the operational scale used for evaluation; only the aggregated loss contribution entering $\mathcal{L}$ is divided by 100 MW.

OSI loss. The stress head output passes through a sigmoid activation so that $\hat{y}^{\text{osi}}$ lies in $[0, 1]$, matching the bounded OSI target constructed in Section 6.1. Stress fit is measured with mean squared error:

$$\mathcal{L}_{\text{osi}} = \frac{1}{B} \sum_{i=1}^B (\hat{y}_i^{\text{osi}} - y_i^{\text{osi}})^2$$

at batch level (equivalently, MSE over scalar OSI predictions in each mini-batch). Squared error on a unit-interval target provides a smooth, well-conditioned regression signal for the graph-level stress scalar and aligns with the continuous OSI formulation adopted over binary or quantised stress labels.

Task weighting and scale balancing. Without rescaling, $\mathcal{L}_{\text{demand}}^{\text{raw}}$ typically attains values on the order of tens of megawatts, whereas $\mathcal{L}_{\text{osi}}$ remains close to zero because OSI targets are normalised to $[0, 1]$. A direct sum of the two terms would therefore allocate almost all gradient magnitude to the demand head, leaving the stress head under-trained despite joint architectural sharing. Dividing $\mathcal{L}_{\text{demand}}^{\text{raw}}$ by 100 MW compresses the demand term to a $\mathcal{O}(1)$ scale appropriate for combination with MSE on OSI. The weight $\lambda_2 = 20$ further elevates the stress contribution so that both task heads receive comparable optimisation pressure while preserving megawatt-scale demand targets. The composite validation score used for early stopping in Section 6.7 applies the same 100 MW scaling to demand error, keeping model selection consistent with the training objective. The chosen λ_2 and normalisation divisor are fixed prior to training and are not tuned on the test partition.

Optimizer and weight decay. Parameters are updated with the Adam optimiser [28] at learning rate $\eta = 5 \times 10^{-4}$ and weight decay $\lambda_{\text{wd}} = 10^{-4}$. Adam provides per-parameter adaptive step sizes that accommodate the heterogeneous gradient magnitudes arising from the parallel spatial and temporal transformer branches and the two task-specific heads. Weight decay supplies mild ℓ_2 regularisation on trainable weights, discouraging overfitting on the finite daily training window without altering the loss formulation explicitly. Optimisation runs for up to 200 epochs, subject to the early-stopping rule in Section 6.7.

Learning-rate schedule. After each training epoch, the learning rate is adjusted by a ReduceLROnPlateau scheduler that monitors validation macro demand MAE. When this metric fails to improve for five consecutive epochs, the learning rate is multiplied by a factor of 0.5. Validation demand MAE is preferred for learning-rate adaptation because it varies smoothly across epochs and supplies a stable optimisation signal on the chronologically ordered validation set; model selection and early stopping nonetheless rely on the composite validation score defined in Section 6.7, so checkpoint retention reflects balanced multi-task performance rather than demand fit alone. The scheduler therefore operates independently of the early-stopping criterion, allowing learning-rate decay to proceed even when the composite score has not yet triggered termination.

TABLE I
DATASET SUMMARY.

Property	Value
Domain	Bangladesh national power grid (9 divisions)
Input window (T)	7 days
Forecast horizon	1 day
Node features	9
Global features	17
Train windows	1,281
Validation windows	263
Test windows	264

TABLE II
FROZEN TRAINING CONFIGURATION (S2 / A6).

Setting	Value
Loss	$\mathcal{L} = \text{Huber}(\text{demand})/100 + \lambda_2 \cdot \text{MSE}(\text{OSI})$
λ_2	20.0
Optimiser	Adam, lr = 5×10^{-4} , wd = 10^{-4}
Batch size	32
Early stopping	Patience 15 (composite val. score)
Parameters	749,058

Gradient clipping. Global gradient-norm clipping with maximum norm 1.0 is applied immediately before each optimiser step. Clipping bounds the ℓ_2 norm of the concatenated parameter gradient vector, mitigating occasional large back-propagation steps that can arise from outlier demand residuals or from attention layers in the graph transformer branch. This stabilises training across epochs without changing the defined loss function.

Numerical precision. Full-precision arithmetic is used throughout training. This methodological choice prioritises numerical stability on the modest daily sample size over the throughput gains of reduced-precision optimisation, which was not required for reliable convergence of the PF-STGT model.

Optimisation rationale. The optimisation design follows three principles aligned with the multi-task formulation. First, robust demand regression (Huber, macro-averaged) respects heterogeneous regional load scales and limits sensitivity to rare extreme demand days. Second, explicit scale balancing (demand loss normalisation and λ_2) ensures that the shared representation learned by PF-STGT encodes stress-relevant structure rather than collapsing to a demand-only solution. Third, conservative numerical tooling (Adam, weight decay, plateau scheduling, gradient clipping, and full-precision arithmetic) prioritises stable convergence on a daily-resolution dataset with fewer than 1,300 training windows. The frozen hyperparameter values, loss coefficients, and early-stopping specification are consolidated in Table II.

IV. EXPERIMENTAL SETUP

A. Experimental Environment

All experiments reported in this study were executed within a single, version-controlled software environment designed to support reproducible training, evaluation, and post-hoc analysis of the PF-STGT framework described in Chapter

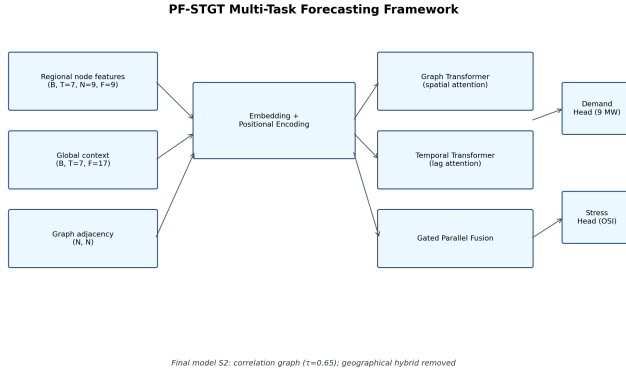


Fig. 1. End-to-end PF-STGT multi-task forecasting framework.

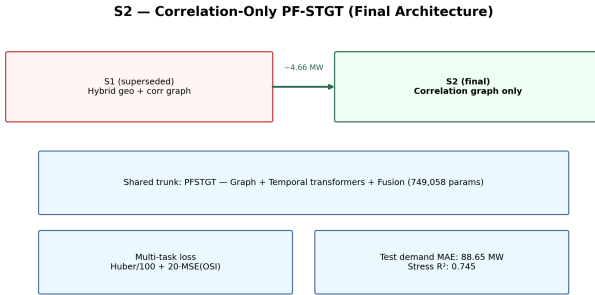


Fig. 2. Architecture freeze: S2 correlation-only graph PF-STGT.

6. The environment couples a pinned Python scientific stack with PyTorch-based deep learning modules, classical machine-learning routines, and explainability tooling. Hardware utilisation follows a portable device-selection policy so that the same codebase can run on CUDA-capable GPUs, Apple Metal accelerators, or CPU fallback without altering model definitions or data pipelines. This subsection documents the compute platform, software dependencies, and reproducibility controls that bound the experimental work; training hyperparameters and evaluation protocols are specified in subsequent subsections.

Compute platform. Deep-learning experiments select the execution device automatically at runtime using the priority order CUDA, Metal Performance Shaders (MPS), then CPU. When an NVIDIA GPU with CUDA support is available, training and inference use it; otherwise, the PyTorch MPS backend is selected if present; if neither accelerator is available, computation proceeds on CPU. This policy is implemented uniformly across training, ablation, benchmark, and explainability scripts. Frozen reference runs for the proposed PF-STGT models—including initial training, ablation studies, and explainability analysis—were executed on the **MPS** backend, as recorded in experiment metadata. Classical benchmark models (linear regression, random forest, and gradient boosting) are trained and evaluated on **CPU** without loading the PyTorch deep-learning stack, isolating lightweight tabular learners from

accelerator-dependent code paths. Reproducibility is anchored on the documented compute backend and pinned software versions used for the frozen experimental programme.

Software environment. Experiments were conducted in a Conda environment named `smartgrid-stgt` with **Python 3.11**, as specified in `environment.yml`. Python package versions are additionally pinned in `requirements.txt` to lock the dependency graph of the publication-freeze version of the repository (25 June 2026). The deep-learning core comprises **PyTorch 2.3.1** [29] and **torchvision 0.18.1**. Graph-aware model components rely on **PyTorch Geometric 2.5.3** [30] and **NetworkX 3.3**. The numerical and tabular data stack includes **NumPy 1.26.4**, **Pandas 2.2.2**, **SciPy 1.13.1**, and **scikit-learn 1.5.0** [31]. Post-hoc explainability modules draw on **SHAP 0.45.1** [18] and **Captum 0.7.0** [32]. Configuration management, logging, and experiment orchestration use **PyYAML 6.0.1**, **Hydra 1.3.2**, **OmegaConf 2.3.0**, and **tqdm 4.66.4**. Visualisation and notebook utilities (**Matplotlib 3.9.0**, **Seaborn 0.13.2**, **JupyterLab 4.2.2**) support diagnostic plotting but are not on the critical path for frozen result generation. **CUDA** is supported by the device-selection policy when a compatible NVIDIA driver and GPU are present; the frozen reference experiments documented in the publication package executed on the **MPS** backend, and no CUDA toolkit version is recorded for those runs.

Reproducibility controls. Three layers of environmental control complement the methodological safeguards in Sections 6.3, 6.7, and 6.8. First, dependency versions are frozen in `requirements.txt` and `environment.yml`, enabling reconstruction of the interpreter and library stack. Second, a fixed random seed (42) governs weight initialisation, data-loader shuffling, and stochastic operations in the reference deep-learning runs, with seed-setting utilities applied consistently across NumPy, Python, and PyTorch. Third, cleaned datasets, engineered feature representations, and graph specifications are verified for consistency against the frozen experimental inputs established in the methodology, ensuring that each run consumes identical data and graph artefacts. All reported results were generated from the publication-freeze version of the repository and are not regenerated during manuscript preparation. Together, the pinned software environment, portable compute policy, and input consistency checks provide a reproducible execution context appropriate for a daily-resolution national grid dataset whose training partition contains fewer than 1,300 sliding-window samples.

Environmental suitability. The chosen environment matches the scale and structure of the study. PyTorch and PyTorch Geometric supply native support for batched graph-tensor operations, transformer blocks, and automatic differentiation required by PF-STGT. The automatic CUDA/MPS/CPU fallback allows development and frozen experimentation on Apple Metal hardware while preserving compatibility with CUDA workstations used in typical deep-learning workflows. Pinned versions of NumPy, Pandas, and scikit-learn ensure that classical baselines and evaluation statistics are computed with the same numerical libraries as the deep-learning pipeline.

CPU execution for classical models avoids unnecessary accelerator overhead on low-parameter learners. The overall configuration prioritises reproducibility and moderate compute requirements over distributed or large-cluster training, which is proportionate to the dataset size and model capacity established in Chapter 6.

B. Baseline Models

Benchmark comparisons evaluate the proposed forecasting framework against seven reference models (B01–B07) [2], [3], [5], [21], [33] spanning classical machine learning, recurrent deep learning, and graph-based spatio-temporal learning. The suite forms a progressive complexity ladder so that performance differences can be attributed to identifiable methodological families. Baselines B01–B06 target Task 1 (regional demand) as the primary comparison axis; B07 alone supports joint Task 1 and Task 2 (OSI) prediction as the historical PF-STGT internal reference. The final proposed model, S2 (correlation-graph PF-STGT), is evaluated separately in the ablation study and is not assigned a B-series identifier.

Classical machine-learning baselines (B01–B03). Three tabular learners provide low-to-moderate complexity reference points that are widely used in load-forecasting literature and require no graph or sequence-specialised architecture.

B01 — Linear Regression represents the simplest multivariate statistical baseline. Inputs consist of the seven-day engineered feature window flattened into a single vector per forecast origin, combining regional node features and national global features. Demand is predicted with an ordinary least-squares multi-output regressor across the nine divisions. Linear regression establishes a transparent lower bound on forecast accuracy and tests whether the engineered feature set alone carries predictive signal without non-linear interactions.

B02 — Random Forest [2] belongs to the ensemble tree family and captures non-linear feature interactions and variable importance through bagged decision trees. Random forest is included because ensemble methods form a dominant cluster in hybrid load-forecasting studies and frequently achieve strong performance on tabular engineered features without explicit temporal or spatial structure. The model is trained under its frozen optimised configuration (Section 7.4).

B03 — XGBoost [3] implements gradient-boosted decision trees with histogram-based training. XGBoost complements random forest within the classical tier by representing the boosting family, which often excels on heterogeneous tabular inputs. Together, B01–B03 isolate the value of non-linearity and ensemble capacity relative to the deep-learning and graph-based tiers. Complete hyperparameter settings are reported in Section 7.4.

For all classical models, node and global tensors are standardised using a scaler fit exclusively on the training partition before regression. Although the primary benchmark comparison focuses on demand forecasting, each classical model also fits a separate scalar regressor for OSI so that stress metrics can be reported on a common evaluation footing; these stress

predictions are secondary and do not define the classical tier’s methodological role.

Deep temporal baselines (B04–B05). Recurrent neural networks test whether sequence modelling without explicit graph structure can capture the temporal dependencies encoded in the seven-day input window.

B04 — LSTM [5] employs a two-layer long short-term memory network with hidden dimension 128 and inter-layer dropout of 0.1. At each time step, regional node features and global features are concatenated into a single input vector; shared recurrent weights process the sequence and a linear head outputs nine regional demand forecasts at the one-day horizon. LSTM is selected because recurrent architectures remain a standard deep-learning baseline in short-horizon energy forecasting and isolate temporal recurrence from graph-based spatial coupling.

B05 — GRU [33] follows the identical input encoding and head structure as B04 but replaces LSTM cells with gated recurrent units. Including both LSTM and GRU controls for cell-type variation within the recurrent family and tests whether gating complexity materially affects demand forecasts on daily windows of length $T = 7$.

Both models are trained under the demand-primary protocol applied to B01–B06. An auxiliary stress head enables comparable OSI evaluation but is not the primary training objective.

Graph-based spatio-temporal baseline (B06). *B06 — T-GCN* [21] is the sole graph neural network in the benchmark suite external to PF-STGT. At each time step, regional node features are projected, aggregated through single-hop graph convolution with the row-normalised hybrid adjacency matrix from Section 6.5, and combined with projected global features before temporal aggregation by a GRU. T-GCN isolates graph convolution plus recurrent pooling without the parallel transformer fusion or multi-task heads of PF-STGT. B06 and B07 share the same hybrid adjacency prior for a controlled comparison against graph-transformer fusion.

Multi-task PF-STGT reference (B07). *B07 — PF-STGT (W20)* is the historical PF-STGT reference configuration used as an internal baseline within the benchmark suite. It implements the parallel-fusion spatio-temporal graph transformer described in Section 6.6 under the W20 multi-task protocol (Section 6.8), with hybrid geographical–correlation graph conditioning. B07 jointly predicts regional demand and graph-level OSI from a shared latent representation and serves as the multi-task graph-transformer reference against which B01–B06 are compared on Task 1; it alone supports primary Task 2 evaluation among the B-series models. S2, the final proposed model (correlation-only graph PF-STGT), is not a B-series benchmark identifier: it is evaluated separately in the ablation study (configuration A6) and supersedes B07 as the publication-facing architecture while retaining the same training protocol documented in Table II.

Fairness protocol. All benchmark models adhere to a common experimental contract designed to prevent information leakage and to ensure that performance differences reflect modelling choices rather than data-handling inconsistencies.

Data partitioning. Every model uses the identical chronological train, validation, and test partitions defined in Section 6.2 and Table I. No model accesses validation or test observations during training except through the prescribed model-selection procedures described below. The first seven rows of each partition are excluded as warm-up, and sliding windows of length $T = 7$ with horizon $H = 1$ are constructed identically across models.

Preprocessing and features. All models read from the same frozen engineered feature store produced by the preprocessing and feature-engineering pipeline in Sections 6.3 and 6.4. Leakage policies established in Section 6.1 apply uniformly: same-day OSI is excluded from inputs when predicting next-day OSI, and normalisation statistics for stress targets are fit on the training partition only. Classical models apply an additional standardisation step to flattened inputs, with scaler parameters fit on training data only.

Feature availability and graph priors. Models that do not use graph structure (B01–B05) consume the same node-level and global feature tensors reduced to tabular or sequential form. Graph-based models (B06, B07) additionally receive the shared hybrid adjacency matrix and, for B07, the derived attention-bias matrix. No baseline is given access to features, graph edges, or targets unavailable to the others at the same forecast origin.

Evaluation metrics. Task 1 performance is assessed with macro-averaged mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) computed per region on the held-out test split and averaged over the nine divisions, as specified in the frozen evaluation protocol. Task 2 metrics (MAE, RMSE, R^2) apply to models that produce OSI forecasts. The test split is evaluated once after model selection; no test-set tuning is permitted.

Hyperparameter tuning. Deep-learning baselines (B04–B06) follow a common optimisation protocol—shared optimiser, learning-rate schedule, early stopping, and batch configuration—described in Section 7.4. Classical models (B01–B03) use frozen library configurations selected under the same validation-based protocol. All hyperparameter values for the reference runs are consolidated in Section 7.4 and Table II. Seed 42 governs the frozen reference experiments.

C. Evaluation Metrics

Model performance is assessed on the held-out test partition defined in Section 6.2 after all training and model-selection decisions have been finalised on the training and validation splits. Evaluation is task-specific: Task 1 (regional evening-peak demand) is the primary comparison axis for all benchmark models in Section 7.2, whereas Task 2 (Operational Stress Index forecasting) is reported for multi-task configurations that produce graph-level OSI outputs. This subsection specifies the metric definitions, aggregation rules, and interpretation principles that govern all quantitative reporting in subsequent results sections. A single metric is insufficient because demand and stress targets differ in scale, units, and operational

meaning; the frozen protocol therefore reports a compact set of complementary error and goodness-of-fit measures for each task.

Evaluation scope. All metrics in this study are computed on the test split containing 264 valid sliding-window forecast origins (Table I). Predictions $\hat{y}^{\text{demand}}$ and $\hat{y}^{\text{osi}}$ are compared against the targets prepared in Sections 6.1 and 6.7 without refitting scalars, graph parameters, or model weights. Validation metrics guide training and hyperparameter selection; test metrics are evaluated once and constitute the primary record for publication.

Task 1 — Regional demand metrics. For each region $r \in \{1, \dots, N\}$ with $N = 9$, let $D_{r,i}$ denote the observed evening-peak demand (MW) and $\hat{D}_{r,i}$ the model prediction on test sample i , where $i = 1, \dots, n$ and n is the number of test windows. Per-region metrics are computed first; national summary scores are obtained by macro-averaging across regions unless stated otherwise.

Mean absolute error (MAE) measures average forecast deviation in megawatts, the native unit of grid demand:

$$\text{MAE}_r = \frac{1}{n} \sum_{i=1}^n |D_{r,i} - \hat{D}_{r,i}|, \quad (3)$$

$$\text{MAE}_{\text{macro}} = \frac{1}{N} \sum_{r=1}^N \text{MAE}_r. \quad (4)$$

MAE is the **primary ranking metric** for Task 1 because it is expressed in directly interpretable physical units, is robust to moderate outliers relative to squared-error measures, and aligns with operator-facing dispatch error reporting.

Root mean squared error (RMSE) penalises large regional forecast errors more heavily than MAE:

$$\text{RMSE}_r = \sqrt{\frac{1}{n} \sum_{i=1}^n (D_{r,i} - \hat{D}_{r,i})^2}, \quad (5)$$

$$\text{RMSE}_{\text{macro}} = \frac{1}{N} \sum_{r=1}^N \text{RMSE}_r. \quad (6)$$

RMSE is reported alongside MAE to indicate whether a model’s error distribution is dominated by occasional large misses, which is relevant for reserve and security assessment.

Mean absolute percentage error (MAPE) expresses relative error as a percentage, supporting comparison with load-forecasting studies that report scale-normalised accuracy:

$$\text{MAPE} = \frac{100}{|\mathcal{V}|} \sum_{(r,i) \in \mathcal{V}} \frac{|D_{r,i} - \hat{D}_{r,i}|}{|D_{r,i}|},$$

where $\mathcal{V}$ is the set of valid region-day pairs with non-negligible observed demand ($|D_{r,i}| > \varepsilon$) to avoid division instability. MAPE is reported for cross-paper comparability but is not used for model selection because it is sensitive to low-demand days and regional scale differences.

Coefficient of determination (R^2) quantifies the proportion of regional demand variance explained by the model on the test split:

$$R_r^2 = 1 - \frac{\sum_{i=1}^n (D_{r,i} - \hat{D}_{r,i})^2}{\sum_{i=1}^n (D_{r,i} - \bar{D}_r)^2}, \quad R_{\text{macro}}^2 = \frac{1}{N} \sum_{r=1}^N R_r^2,$$

where $\bar{D}_r$ is the mean observed demand in region r over the test period. **Macro** R^2 is defined as the unweighted mean of per-region R_r^2 values, not as a single pooled R^2 over all region-day pairs. This aggregation treats each division equally and prevents high-load regions from dominating the variance-explained statistic. R^2 complements MAE by indicating how well a model tracks regional demand variability; the two metrics need not rank models identically when prediction variance differs from observed variance.

In addition to macro summaries, **Dhaka** division MAE is reported separately. Dhaka carries the largest share of national evening-peak load and is the principal demand centre for national transmission and day-ahead dispatch planning; isolating its forecast error therefore provides an operationally focused diagnostic beyond the unweighted macro average. Per-region MAE values for all nine divisions are available in the frozen results package for further analysis.

Task 2 — Operational stress metrics. Task 2 evaluates graph-level OSI predictions $\hat{y}_i^{\text{osi}}$ against targets $y_i^{\text{osi}} \in [0, 1]$ on the same test windows. Because OSI is a bounded scalar index rather than a nine-dimensional regional vector, metrics are computed directly over the n test samples without regional macro-averaging:

$$\begin{aligned} \text{MAE}_{\text{osi}} &= \frac{1}{n} \sum_{i=1}^n |y_i^{\text{osi}} - \hat{y}_i^{\text{osi}}|, \\ \text{RMSE}_{\text{osi}} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i^{\text{osi}} - \hat{y}_i^{\text{osi}})^2}, \\ R_{\text{osi}}^2 &= 1 - \frac{\sum_{i=1}^n (y_i^{\text{osi}} - \hat{y}_i^{\text{osi}})^2}{\sum_{i=1}^n (y_i^{\text{osi}} - \bar{y}^{\text{osi}})^2}. \end{aligned}$$

MAE_{osi} is the primary stress error measure because OSI lies on a unit interval and absolute deviation is directly interpretable as forecast bias on the normalised stress scale. RMSE_{osi} highlights large stress-prediction failures that may correspond to under-predicted operational pressure. R_{osi}^2 indicates whether the model captures day-to-day stress variability beyond the test-set mean. **Pearson correlation** $r(\hat{y}_i^{\text{osi}}, y_i^{\text{osi}})$ is reported only as a supplementary descriptive statistic for multi-task models and is not used for model ranking or selection. Simple reference forecasts—persistence ($\hat{y}_i^{\text{osi}} = y_{i-1}^{\text{osi}}$) and the training-set median—provide contextual baselines for interpreting learned stress predictions; they are not benchmark models defined in Section 7.2.

Macro averaging and metric interpretation. Macro-averaging for Task 1 assigns equal weight to each administrative division, consistent with the node-level forecasting formulation in Section 6.1 and preventing a single high-load region from determining the national leaderboard score. This choice reflects the study’s objective of equitable regional forecast quality across Bangladesh’s nine divisions rather than optimising exclusively for the national aggregate. For Task 2, no macro average is required because supervision is graph-level.

Multiple metrics are reported jointly because each captures a distinct facet of forecast quality: MAE and RMSE quantify magnitude of error in physical or normalised units; MAPE supports literature comparison; and R^2 assesses variance explanation. Relying on a single score—particularly R^2 alone—can obscure models that achieve low absolute error but under-represent temporal variability, or vice versa. The frozen protocol designates **macro demand MAE on the test split** as the sole primary metric for ranking Task 1 models; all other Task 1 and Task 2 measures are secondary and interpretive. Formal inferential comparison of paired forecast errors, including Wilcoxon signed-rank testing [34] and Bonferroni correction [35], is specified in Section 7.5 (Statistical Testing); numerical outcomes are reported in Section 8.

D. Hyperparameter Configuration

The evaluation metrics in Section 7.3 are computed under a fixed hyperparameter regime established before the publication freeze and held constant across all reference runs. Hyperparameters were selected on the validation split only; the test partition was not used for tuning. A fixed random seed (42) governs all stochastic components in the frozen reference experiments. This subsection records the training and model-capacity settings for the proposed PF-STGT framework (S2 / ablation A6), the deep-learning baselines (B04–B06), and the classical baselines (B01–B03). Table II provides the authoritative summary for the final proposed model; the narrative below supplements Table II with architecture-capacity parameters and baseline-specific settings that fall outside the table.

Shared windowing parameters. All models share the input sequence length $T = 7$ days and one-day forecast horizon $H = 1$, consistent with Sections 6.1 and 6.2. These values are fixed by the dataset design and are not treated as tunable hyperparameters.

Proposed PF-STGT configuration (S2 / A6). The final proposed model uses the W20 multi-task training protocol with correlation-only graph conditioning. Table II consolidates the optimisation settings: **Adam** [28] optimiser with learning rate $\eta = 5 \times 10^{-4}$ and weight decay $\lambda_{\text{wd}} = 10^{-4}$; mini-batch size $B = 32$; maximum training duration of 200 epochs subject to early stopping with **patience 15** on the balanced validation score $0.7 \cdot (\text{MAE}_{\text{demand}}^{\text{val}}/100) + 0.3 \cdot \text{MAE}_{\text{osi}}^{\text{val}}$; **ReduceLROnPlateau** scheduling on validation macro demand MAE (multiplicative factor 0.5, scheduler patience 5); and **gradient-norm clipping** at maximum norm 1.0. Multi-task

loss weighting uses stress coefficient $\lambda_2 = 20.0$ with **demand-loss scaling** that divides the Huber demand term by 100 MW before combination with OSI mean squared error, as specified in Section 6.8.

Architectural capacity hyperparameters for PF-STGT were fixed during architecture selection: hidden dimension $d_{\text{model}} = 128$, four attention heads, two spatial transformer layers and two temporal transformer layers, feed-forward inner dimension 256, and **dropout** rate 0.1 in both spatial and temporal transformer branches. The stress-prediction head uses an internal hidden dimension of 128. These capacity choices balance representational power against the approximately 1,281 training windows available after warm-up exclusion.

Deep-learning baseline configuration (B04–B06). Recurrent baselines B04 (LSTM) and B05 (GRU) employ a two-layer sequence encoder with hidden dimension 128 and inter-layer dropout of 0.1; graph baseline B06 (T-GCN) uses hidden dimension 64 in its convolutional and temporal blocks. Deep baselines are trained with **AdamW** [36] (learning rate 5×10^{-4} , weight decay 10^{-4}), batch size 32, gradient-norm clipping at 1.0, ReduceLROnPlateau scheduling on validation macro demand MAE (factor 0.5, patience 5), and early stopping with patience 15 on validation demand MAE. Training runs for up to 60 epochs in the frozen benchmark experiment, retaining the checkpoint with lowest validation demand MAE. Demand fit uses macro-averaged Huber loss during training; these baselines follow the demand-primary protocol described in Section 7.2.

Classical baseline configuration (B01–B03). Classical models operate on flattened seven-day feature vectors with **standardisation** parameters fit exclusively on the training partition. B01 (linear regression) uses ordinary least-squares multi-output regression for demand without additional regularisation hyperparameters. B02 (random forest) and B03 (XGBoost) use frozen ensemble configurations selected during validation-based hyperparameter tuning: 300 decision trees for random forest, and for XGBoost 300 estimators with maximum tree depth 6 and learning rate 0.1. Classical fits are deterministic given the fixed seed and feature matrix; no iterative early stopping applies.

Configuration suitability. The frozen hyperparameter set is proportionate to the daily-resolution dataset and the multi-task objective. Batch size 32 and learning rate 5×10^{-4} were among the candidates evaluated during validation-based hyperparameter tuning and provide stable convergence across PF-STGT and deep baseline architectures. Capacity parameters ($d_{\text{model}} = 128$, two transformer layers per branch) avoid over-parameterisation relative to the training-window count while preserving the spatial and temporal representational depth required by the graph-transformer design. Multi-task settings ($\lambda_2 = 20$, demand scaling by 100 MW) are fixed from the W20 protocol established before ablation and benchmark reporting. Gradient clipping and weight decay supply mild regularisation without altering the loss definitions in Section 6.8. Classical ensemble configurations were chosen to represent strong tabular competitors at moderate computational cost on

CPU. Together, Table II and the settings above define the complete reproducible hyperparameter record for all experiments reported in this study.

E. Statistical Testing

The hyperparameter configuration in Section 7.4 fixes the models under comparison; this subsection specifies how paired forecast errors on the held-out test split are tested for systematic differences. Inferential analysis applies exclusively to **Task 1** and uses the primary evaluation metric defined in Section 7.3—macro-averaged demand MAE in megawatts—computed independently for each test-window forecast origin. Comparisons are organised into three pre-specified families: benchmark models (reference PF-STGT versus B01–B06), ablation variants (reference A1 versus A2–A6), and architecture-simplification variants (reference S1 versus S2–S4). Numerical test outcomes are reported in Section 8; the present subsection defines the protocol only.

Paired comparison unit. For each test window $i = 1, \dots, n$ with $n = 264$, every model produces a vector of nine regional demand forecasts. The paired statistic is per-window macro MAE:

$$\text{MAE}_i = \frac{1}{N} \sum_{r=1}^N |D_{r,i} - \hat{D}_{r,i}|,$$

with $N = 9$. Because all models are evaluated on the identical set of forecast origins, differences $\Delta \text{MAE}_i = \text{MAE}_i^{(\text{ref})} - \text{MAE}_i^{(\text{cand})}$ form paired observations across the same test days. This pairing controls for day-specific grid conditions—calendar effects, weather, and operational events—that would confound unpaired comparisons.

Wilcoxon signed-rank test. The primary inferential test is the **paired Wilcoxon signed-rank test** [34] applied to $\{\Delta \text{MAE}_i\}$. The null hypothesis is that the median paired difference is zero. A non-parametric test is appropriate because per-window forecast errors are typically skewed, may contain occasional large deviations on high-load or stressed days, and are not assumed to follow a Gaussian distribution. The signed-rank procedure uses the magnitudes of non-zero differences while respecting their signs, making it robust to outliers relative to paired t -tests. **Two-sided** p -values are reported as the default for general model comparisons. **One-sided** p -values are computed only where a pre-registered directional hypothesis specifies that the reference model should improve upon a particular baseline; Bonferroni-adjusted significance for such claims uses the appropriate one-sided probability.

Multiple-comparison correction. Each comparison family involves several simultaneous hypothesis tests. **Bonferroni correction** [35] controls the family-wise error rate by adjusting the nominal significance level. For the benchmark family (six pairwise comparisons of the reference PF-STGT model against B01–B06), the adjusted threshold is $\alpha_B = 0.05/6 \approx 0.0083$. For the ablation family (five comparisons of A1 against A2–A6), $\alpha_B = 0.05/5 = 0.01$. The architecture-simplification family applies the same $\alpha_B = 0.01$ criterion for comparisons

Figure 3 — Training Curves (frozen historical run)

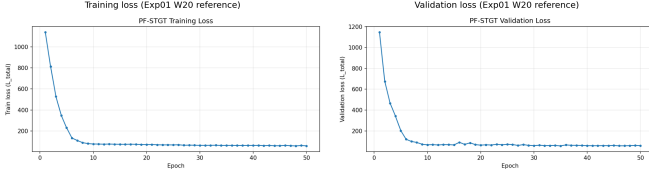


Fig. 3. Training and validation loss curves (W20 protocol reference).

against S1. A comparison is declared statistically significant only when the relevant one-sided or two-sided p -value falls below the family-specific Bonferroni-adjusted threshold and the direction of the median paired difference is consistent with the stated hypothesis.

Effect size and uncertainty quantification. Statistical significance is supplemented by **effect size** and **bootstrap confidence intervals** to support interpretation. For benchmark comparisons, **Cohen’s d** [37] is computed from the paired differences as the mean Δ MAE divided by the sample standard deviation of Δ MAE, with $|d| \geq 0.5$ treated as a medium effect in reporting guidelines. Cohen’s d is reported as a standardised descriptive measure of practical effect magnitude that complements the non-parametric Wilcoxon test; it does not replace a rank-based effect-size statistic but aids interpretation of the megawatt-scale paired differences on a common scale. **Bootstrap 95% confidence intervals** [38] on the mean paired MAE difference are obtained by resampling test windows with replacement (2,000 resamples, fixed random seed) and taking the 2.5th and 97.5th percentiles of the resampled mean differences. Bootstrap intervals provide distribution-free uncertainty estimates for the mean paired improvement or degradation without assuming normality of the error differences.

Statistical significance versus practical importance. Rejecting the null hypothesis indicates that a model’s per-window macro MAE differs systematically from its paired reference on the test split; it does not, by itself, establish operational value. **Practical importance** is assessed from the magnitude of paired MAE differences expressed in megawatts, from bootstrap confidence intervals that exclude zero, and from effect-size conventions. A statistically significant improvement of very small megawatt magnitude may be inconsequential for national dispatch planning, whereas a moderate MAE reduction with a medium Cohen’s d and a bootstrap interval entirely below zero provides stronger evidence of meaningful gain. Conversely, a non-significant test does not prove equivalence; insufficient test-window count or high day-to-day variance may limit power to detect modest but operationally relevant differences. Section 8 presents these outputs in the order encountered by the reader: raw MAE metrics first, followed by bootstrap confidence intervals, then p -values with Bonferroni-adjusted thresholds, and finally Cohen’s d where applicable—with macro demand MAE remaining the primary ranking criterion from Section 7.3.

V. RESULTS

A. Overall Benchmark Performance

1) Overall benchmark summary: The benchmark evaluation compares the proposed correlation-graph PF-STGT model (S2) against seven reference models (B01–B07) on the held-out test split comprising 264 forecast windows. Table III consolidates macro-averaged demand metrics over nine administrative divisions together with graph-level Operational Stress Index (OSI) metrics where available. The primary ranking criterion is test macro demand mean absolute error (MAE) in megawatts.

Among all entries in Table III, S2 attains the lowest demand MAE at 88.65 MW, followed by the historical hybrid-graph PF-STGT reference B07 at 93.31 MW and the random forest baseline B02 at 97.03 MW. At the opposite end, the temporal graph convolutional baseline B06 records the highest demand MAE at 257.21 MW, with linear regression B01 close behind at 247.79 MW. This separation indicates that the proposed multi-task graph-transformer configuration occupies the leading position on the primary metric within the frozen benchmark suite, while the recurrent baselines B04 (LSTM) and B05 (GRU) and the graph-convolution baseline B06 (T-GCN) [21] remain far above the competitive tier on megawatt MAE. For day-ahead national dispatch planning, the relevant takeaway is that the final proposed model delivers the smallest average absolute regional forecast error among all benchmarked learners on the 2024 test period.

2) Demand forecasting performance: The lowest macro demand MAE in Table III belongs to S2 (88.65 MW; RMSE 127.29 MW; MAPE 6.55%). B07, the prior PF-STGT W20 hybrid reference, records 93.31 MW MAE with RMSE 128.81 MW and MAPE 6.76%. The strongest classical competitor, random forest B02 [2], reaches 97.03 MW MAE with RMSE 156.99 MW, while gradient boosting B03 [3] yields 109.73 MW MAE. These three models form a distinct upper tier in which graph-aware or ensemble tabular methods outperform the remaining candidates by a wide margin.

Recurrent baselines without explicit graph structure perform poorly: B04 (LSTM) [5] and B05 (GRU) [33] achieve demand MAE of 237.03 MW and 233.48 MW respectively, with negative macro R^2 values (-0.242 and -0.201). The graph-based but non-transformer baseline B06 (T-GCN) records the worst demand MAE at 257.21 MW and macro R^2 of -0.483. Linear regression B01 remains far above the competitive tier at 247.79 MW MAE and MAPE 17.32%. The pattern shows that graph conditioning alone, in the form of a shallow spatio-temporal GCN [6], does not translate into accurate regional forecasts on this dataset, whereas the PF-STGT family—whether hybrid (B07) or correlation-only (S2)—consistently outperforms both classical ensembles and standalone recurrent encoders.

A notable secondary observation concerns demand R^2 . Random forest B02 reports macro R^2 of 0.984, exceeding S2 (0.684) and B07 (0.674). R^2 and MAE quantify different aspects of predictive performance—variance explained rela-

tive to the regional mean versus mean absolute megawatt deviation—and macro demand MAE remains the predefined primary ranking metric in Table III. On that criterion, S2 and B07 rank ahead of B02 despite the higher classical R^2 .

S2 improves upon B07 by 4.66 MW in mean demand MAE (88.65 versus 93.31 MW), a reduction of approximately 5.0% relative to the historical reference. Within the multi-task models reported in Table III, S2 thus represents the stronger demand forecaster. In absolute terms, reductions of this magnitude—and the 8.38 MW point-estimate gap between S2 and random forest B02—represent measurable decreases in mean absolute megawatt forecast error on the test split, without extending the benchmark evidence to quantified downstream operational or economic effects. The practical implication is that adopting the correlation-only graph variant of PF-STGT yields modest but consistent megawatt-level gains over the earlier hybrid-graph configuration without relinquishing joint stress prediction.

3) *Operational Stress prediction*: Task 2 performance is reported only for models that produce graph-level OSI forecasts. In Table III, S2 achieves stress MAE of 0.0371 and stress R^2 of 0.745—the highest OSI explanatory power among all benchmarked models. B07 records stress MAE of 0.0499 and stress R^2 of 0.585. Classical and recurrent baselines that supply auxiliary stress outputs perform worse: B02 stress R^2 is 0.555 (MAE 0.0481), B03 is 0.525 (MAE 0.0497), and B04–B06 yield negative stress R^2 values ranging from -0.191 to -0.304 with stress MAE between 0.086 and 0.089.

The proposed model therefore delivers both the lowest OSI absolute error and the strongest variance explanation on the test split. Relative to B07, S2 reduces stress MAE from 0.0499 to 0.0371 and raises stress R^2 by 0.160 absolute (0.585 to 0.745). For grid operators monitoring operational pressure alongside load, this indicates that the final multi-task configuration not only improves demand forecasts but also provides a substantially more informative day-ahead stress signal than the historical PF-STGT reference and than any stress-capable baseline in the benchmark suite.

4) *Statistical evidence*: Paired Wilcoxon signed-rank tests [34] on per-window macro demand MAE ($n = 264$) provide inferential support for the benchmark rankings in Table IV. With B07 as the reference and Bonferroni-adjusted significance threshold [35] $\alpha = 0.0083$ across six comparisons, B07 achieves statistically significant lower demand MAE than every baseline B01–B06. Median paired improvements range from 4.92 MW against random forest B02 to 160.66 MW against T-GCN B06. Bootstrap 95% confidence intervals [38] on mean paired MAE differences exclude zero for comparisons against B01, B03, B04, B05, and B06; the interval for B07 versus B02 is [-8.87, 2.62] MW, which spans zero despite a significant Wilcoxon result ($p = 0.00135$, Cohen’s d [37] = -0.077).

Direct inferential evidence for S2 versus the B07 reference is available from the paired Wilcoxon comparison on the same test windows: median Δ MAE = -5.43 MW, two-sided $p = 5.5 \times 10^{-5}$, bootstrap 95% CI [-7.17, -2.16] MW. This interval

excludes zero and corroborates the point estimate improvement seen in Table III. No formal paired Wilcoxon test of S2 against random forest B02 was conducted in the benchmark experiment; the point estimate advantage of 8.38 MW (97.03 - 88.65) should be interpreted alongside the significant B07-versus-B02 result rather than as an independently tested claim.

Taken together, the statistical record supports systematic PF-STGT superiority over classical, recurrent, and GCN baselines on megawatt MAE, confirms a significant incremental gain of the final S2 configuration over the B07 reference, and cautions that the B07-versus-random forest improvement—while statistically significant—carries a small standardized effect size and a bootstrap interval compatible with near-zero mean improvement. Dispatch applications should weight megawatt point estimates and confidence bounds alongside p -values when judging operational relevance.

5) *Overall interpretation*: The frozen benchmark results establish S2 as the leading multi-task configuration in Table III on both reported tasks: lowest macro demand MAE (88.65 MW) and highest stress R^2 (0.745) on 264 held-out test windows. The PF-STGT family consistently outperforms the standalone LSTM, GRU, and T-GCN architectures, whose demand errors exceed 230 MW and whose stress predictions fail to outperform the test-set mean. Classical ensembles remain competitive tabular benchmarks—random forest in particular—but do not match the megawatt accuracy of the proposed graph-transformer model despite higher reported macro R^2 .

The principal strength of the benchmark outcome is the joint delivery of accurate regional demand forecasts and informative OSI predictions from a single multi-task model, with statistically supported improvements over both external baselines and the prior hybrid-graph PF-STGT reference. The principal limitation visible at this stage is that demand MAE gains over the strongest non-PF-STGT baseline (random forest) are modest in absolute terms and accompanied by a wide bootstrap confidence interval in the B07-versus-B02 test, and that classical models can exhibit favourable R^2 without corresponding MAE leadership. These results validate the proposed framework as the preferred multi-task forecasting configuration in Table III while indicating that absolute megawatt improvements over well-tuned ensemble methods, though consistent, should be communicated with appropriate statistical qualification rather than as unqualified dominance across every secondary metric.

B. Regional Forecast Analysis

1) *Macro context and regional scope*: Section 8.1 established model rankings using macro-averaged demand MAE over nine administrative divisions on 264 test windows. Fig. 4 visualises that macro comparison, with S2 attaining the lowest bar at 88.65 MW ahead of B07 (93.31 MW), B02 (97.03 MW), B03 (109.73 MW), and B06 (257.21 MW) as reported in Table III. Because macro MAE assigns equal weight to each division, the chart summarises national leaderboard performance but does not reveal how forecast error is distributed across regions with markedly different load scales.

Division-level demand MAE for S2 is reported in the supplementary ablation outputs; per-region MAE and R^2 for B07 and B02 appear in Table S1. Together with Fig. 4 and Table III, these sources permit a disaggregated reading of the macro results without altering the rankings established in Section 8.1. For operators, the practical value of this regional view is to identify where absolute megawatt errors concentrate and whether macro gains reflect uniform improvement or selective division-level shifts.

2) *Regional demand performance*: Under S2, per-region test MAE ranges from 32.73 MW (Barishal) to 293.98 MW (Dhaka), with the full nine-division profile tabulated in the supplementary ablation outputs. The three lowest-MAE divisions—Barishal, Sylhet, and Rangpur—each remain below 55 MW, whereas Dhaka alone exceeds 290 MW and Khulna exceeds 88 MW. At the macro level, S2 records demand RMSE of 127.29 MW, demand MAPE of 6.55%, and macro demand R^2 of 0.684 on the test split.

Relative to B07 (Table S1), peripheral accuracy is broadly comparable but not uniform: S2 records slightly higher MAE in Barishal, Chattogram, Cumilla, Mymensingh, and Sylhet, while reducing MAE in Khulna (88.62 versus 118.72 MW), Rajshahi (76.25 versus 85.70 MW), Rangpur (53.97 versus 59.77 MW), and Dhaka (293.98 versus 299.78 MW). Against random forest B02, S2 achieves lower Dhaka and Khulna MAE while trailing B02 in Barishal and several other divisions by small margins. The interpretation is that S2's macro leadership in Section 8.1 arises from a mixture of division-level gains and minor regressions rather than from uniform error reduction everywhere. Planners should therefore treat the macro bar in Fig. 4 as an equal-weight summary while consulting Table S1 when prioritising transmission corridors or load centres with the largest absolute residuals.

3) *Regional consistency and variability*: Per-region MAE under S2 exhibits substantial cross-division spread: the ratio between Dhaka MAE (293.98 MW) and Barishal MAE (32.73 MW) exceeds ninefold on the test split. B07 and B02 display a similar pattern in Table S1, with Dhaka MAE of 299.78 MW and 311.59 MW respectively against peripheral MAE values generally below 90 MW. This consistency of error concentration across model families indicates that regional heterogeneity in absolute megawatt error is a persistent feature of the test period rather than an artefact of a single architecture.

Comparing S2 with B07 division by division, the largest improvements occur in Khulna (-30.10 MW), Rajshahi (-9.45 MW), Dhaka (-5.80 MW), and Rangpur (-5.80 MW), whereas S2 records slightly higher MAE than B07 in five peripheral divisions, each by less than 5 MW in absolute terms. The underlying cause of this asymmetric pattern is not established by the benchmark outputs alone and would require further investigation beyond the scope of the present results. The practical implication is that the macro MAE gain of 4.66 MW (S2 versus B07) aggregates offsetting regional effects; deployment reviews should examine division-level metrics alongside the national summary in Fig. 4.

Per-region R^2 values in Table S1 further illustrate metric

divergence. For B07, division-level R^2 ranges from 0.4808 (Barishal) to 0.8270 (Rajshahi); for B02, Rajshahi reaches 0.8274 while Dhaka records 0.6666 despite B02's higher Dhaka MAE than S2. R^2 and MAE therefore need not agree at the division level, consistent with the macro-level distinction noted in Section 8.1.2. Forecast quality assessments that rely on a single metric may reach different conclusions depending on whether absolute megawatt deviation or variance explanation is emphasised.

4) *Dhaka-specific performance*: Dhaka carries the largest evening-peak load among the nine divisions and accounts for approximately 35.7% of mean national regional demand in the exploratory data analysis. On the test split, Dhaka also exhibits the highest demand variability among audited regions, with B07 recording an observed demand standard deviation of 670.15 MW in Table S1—far exceeding peripheral divisions. Dhaka MAE under S2 is 293.98 MW, compared with 299.78 MW for B07 and 311.59 MW for B02 (Table S1); the supplementary audit reports Dhaka R^2 of 0.5744 for B07 and 0.6666 for B02 on the same split.

S2 therefore achieves the lowest audited Dhaka MAE among the three models with published division-level tables, reducing absolute Dhaka error by 5.80 MW relative to B07 and 17.61 MW relative to B02. Because Dhaka's load scale dominates national totals, even modest percentage improvements translate into larger absolute megawatt reductions than comparable percentage gains in low-load divisions. Nevertheless, Dhaka MAE remains an order of magnitude above peripheral MAE under S2, so the principal national load centre continues to account for the largest share of forecast error in megawatt terms. Day-ahead planning for the capital region should continue to treat Dhaka as a separately reported diagnostic rather than relying solely on macro-averaged scores.

5) *Regional strengths and limitations*: The strongest regional performance under S2 occurs in Barishal, Sylhet, and Rangpur, where MAE remains below 55 MW, and in Mymensingh (59.22 MW). These divisions contribute relatively little to the macro numerator despite receiving equal weight in the macro average. In the present benchmark, S2 also delivers its largest MAE reductions relative to B07 in Khulna and Rajshahi, indicating that the proposed model improves selected mid-scale divisions with material evening-peak load without registering worse MAE in every peripheral node.

The principal regional limitation is the persistent concentration of absolute error in Dhaka, where S2 MAE (293.98 MW) still far exceeds all other divisions. Secondary limitations include small MAE regressions versus B07 in five divisions—Barishal, Chattogram, Cumilla, Mymensingh, and Sylhet—each below 5 MW in absolute terms, and the absence of a published per-region R^2 table for S2 in the supplementary materials, which restricts division-level variance diagnostics for the proposed model to MAE and the macro R^2 summary (0.684). Regional patterns beyond the reported MAE and R^2 values—such as seasonal or event-driven error clustering—are not characterised in the benchmark artefacts and are deferred to dedicated error-analysis reporting. Overall, the regional

evidence supports S2 as a consistent multi-division forecaster with operationally meaningful Dhaka gains and stable peripheral accuracy, while highlighting that equal-weight macro summaries in Fig. 4 can understate the national planning significance of the largest load centre.

C. Operational Stress (OSI) Analysis

1) *Scope and evaluation context:* Section 8.1 introduced joint demand and OSI rankings from Table III; this subsection examines Task 2 in greater depth. Operational Stress Index prediction is evaluated on the same held-out test split of 264 forecast windows used for demand benchmarking, with graph-level OSI reported as mean absolute error (MAE) and coefficient of determination (R^2) on the scalar stress target in $[0, 1]$. Only models that emit an OSI forecast appear in the stress columns of Table III; the demand-only configuration A4 in Table V is excluded from OSI comparison because it does not produce a stress output.

The frozen evaluation record therefore permits two complementary readings: an external benchmark comparison of S2 against stress-capable baselines B01–B07 (Table III), and an internal configuration comparison across the multi-task variants tabulated in Table V. Fig. 5 visualises test-set demand MAE for the frozen variants in Table V and is referenced here alongside Table V’s stress columns to characterise how demand and stress accuracy co-vary across configurations. For operators who require a day-ahead stress signal alongside regional load forecasts, the relevant question is whether the final proposed model delivers the strongest joint stress evidence within the frozen suite rather than optimising either task in isolation.

2) *OSI benchmark comparison:* Among all stress-capable entries in Table III, S2 attains the lowest OSI MAE at 0.0371 and the highest OSI R^2 at 0.745 on the 264-window test split. The historical PF-STGT reference B07 records OSI MAE of 0.0499 and OSI R^2 of 0.585. Classical ensemble baselines that supply auxiliary stress outputs remain above S2 on both metrics: random forest B02 achieves OSI MAE of 0.0481 with R^2 of 0.555, and gradient boosting B03 records OSI MAE of 0.0497 with R^2 of 0.525. Recurrent and graph-convolution baselines B04–B06 yield OSI MAE between 0.0861 and 0.0891 with negative stress R^2 values ranging from -0.191 to -0.304, indicating that their stress predictions fail to outperform the test-set mean. Linear regression B01 records the largest OSI MAE at 0.1074 and stress R^2 of -1.824.

Relative to B07, S2 reduces OSI MAE by 0.0128 absolute (0.0499 to 0.0371, approximately 25.7%) and raises stress R^2 by 0.160 absolute (0.585 to 0.745). Relative to the strongest classical stress competitor in Table III, random forest B02, S2 improves OSI MAE by 0.0110 (0.0481 to 0.0371) and stress R^2 by 0.190 (0.555 to 0.745). The benchmark pattern shows that the proposed multi-task graph-transformer configuration occupies the leading position on both reported OSI metrics, while shallow recurrent and T-GCN architectures do not yield informative day-ahead stress forecasts on this split. Grid operators comparing stress-capable models should treat S2

as the preferred OSI forecaster within the frozen external benchmark, with the caveat that classical baselines remain closer on absolute OSI MAE than on variance explanation.

3) *Configuration-level OSI performance:* Table V reports OSI metrics for the frozen multi-task configuration suite evaluated on the identical test split. S2 (variant A6) records OSI MAE of 0.0371 and OSI R^2 of 0.745. The historical hybrid-graph reference A1 (equivalent to B07) records OSI MAE of 0.0499 and OSI R^2 of 0.585. Variants A2 (no graph) and A3 (no transformer) each achieve OSI MAE of 0.0405 and OSI R^2 of 0.701—lower absolute error and higher R^2 than A1 but still behind A6 on both stress metrics. Variant A5 (geographical graph only) attains the lowest OSI MAE among multi-task configurations at 0.0340 and the highest stress R^2 at 0.764, while recording demand MAE of 97.98 MW. Variant A4 (single-task) achieves the lowest demand MAE in the suite at 86.89 MW but provides no OSI output.

Fig. 5 orders the same configurations by test demand MAE: A4 at 86.89 MW, A6 (S2) at 88.65 MW, A3 at 92.64 MW, A1 at 93.31 MW, A2 at 93.93 MW, and A5 at 97.98 MW. Read together with Table V, the figure and table show that lower demand MAE does not imply superior OSI accuracy across configurations. A5 leads on both OSI metrics yet records the highest demand MAE among the multi-task variants shown in Fig. 5, whereas A6 combines the second-lowest demand bar in that chart with the second-best OSI MAE and R^2 in Table V. The benchmark artefacts demonstrate this demand–stress divergence but do not establish the underlying mechanism responsible for variant-level trade-offs. For joint day-ahead planning, S2 represents the configuration that simultaneously delivers the lowest multi-task demand MAE among OSI-producing variants in Fig. 5 and the strongest OSI R^2 in the external benchmark of Table III, accepting that A5 attains marginally lower OSI MAE and higher stress R^2 at the cost of 9.33 MW higher demand error relative to S2.

4) *Statistical evidence and reporting limits:* Formal inferential testing in the frozen statistical record is defined on per-window macro **demand** MAE rather than on OSI residuals. The paired Wilcoxon comparison of A1 versus A6 (S2) reports median demand Δ MAE of -5.43 MW, two-sided $p = 5.5 \times 10^{-5}$, and bootstrap 95% confidence interval $[-7.17, -2.16]$ MW on the test split (Table V; statistical summary). That interval excludes zero and corroborates the demand improvement of S2 over the historical reference; no corresponding Wilcoxon test, p-value, or bootstrap confidence interval for OSI MAE or OSI R^2 was computed in Experiments 02 or 03.

Stress improvement from A1 to A6 is therefore supported only by point estimates: OSI MAE decreases from 0.0499 to 0.0371 and OSI R^2 increases from 0.585 to 0.745. Benchmark comparisons of S2 against classical baselines on OSI similarly rest on Table III point estimates without paired stress hypothesis tests. Wilcoxon results for B07 versus B01–B06 in Table IV establish significant demand MAE separation for the PF-STGT reference but do not extend to the stress task. The interpretation is that OSI performance claims in this section should be communicated as consistent point-

estimate advantages on the frozen test split, while demand-side inferential evidence provides indirect support that the final configuration improves upon the historical reference on the primary optimisation objective without substituting for stress-specific significance testing.

5) *Practical implications and limitations:* The frozen OSI evidence indicates that S2 delivers graph-level stress forecasts with MAE below 0.04 and R^2 of 0.745, explaining approximately three-quarters of test-set OSI variance while also leading the demand benchmark in Section 8.1. Operators integrating a single multi-task model for next-day regional load and national stress monitoring can rely on S2 as the configuration with the strongest joint benchmark credentials in Tables 3 and 5, rather than deploying a classical baseline that trails on stress R^2 (B02: 0.555; B03: 0.525) or a recurrent architecture whose stress predictions are uninformative relative to the mean.

Three limitations follow directly from the reported artefacts. First, no stress-specific confidence intervals or multiple-comparison-corrected tests are available; OSI gains over B07 and classical baselines should be qualified accordingly. Second, configuration A5 achieves lower OSI MAE (0.0340) and higher stress R^2 (0.764) than S2 but at materially higher demand MAE (97.98 versus 88.65 MW), so stress-optimal and demand-optimal configurations diverge within the frozen suite. Third, variant A4 establishes a demand-only performance bound of 86.89 MW—1.76 MW below S2—without furnishing any OSI forecast, confirming an explicit demand–stress Pareto trade-off rather than simultaneous optimality on both tasks. The benchmark demonstrates these patterns on the 2024 test period but does not quantify downstream operational or economic consequences of OSI forecast error; deployment decisions should treat the reported MAE and R^2 values as accuracy diagnostics to be monitored alongside division-level demand metrics from Section 8.2.

D. Ablation Study

1) *Ablation scope and reference configuration:* The frozen ablation programme evaluates six PF-STGT configurations (A1–A6) on the held-out test split of 264 forecast windows, with A1 serving as the reference: hybrid geographical–correlation graph, full transformer trunk, and multi-task demand–OSI learning under the W20 protocol. Table V consolidates test-set macro demand MAE, demand R^2 , stress MAE, and stress R^2 for each variant; Fig. 5 visualises the same configurations ranked by demand MAE. Together, the table and figure permit component-wise comparison against A1 without rerunning experiments.

A1 (equivalent to benchmark reference B07) records demand MAE of 93.31 MW, demand R^2 of 0.674, stress MAE of 0.0499, and stress R^2 of 0.585. All subsequent ablation contrasts in this section are interpreted relative to that reference. The design implication is that any architectural change should be judged against this frozen multi-task baseline rather than against external baselines alone, because A1 encodes the

original hybrid-graph PF-STGT configuration from which S2 was derived.

2) *Graph contribution:* Graph topology is isolated by three variants: A2 removes graph conditioning entirely while retaining the hybrid multi-task trunk; A5 replaces the hybrid adjacency with a geographical graph only; and A6 (S2) replaces it with a correlation-only graph ($\tau = 0.65$). On demand MAE, A2 records 93.93 MW—0.62 MW above A1—whereas A5 records 97.98 MW, the highest demand error among the six configurations. A6 achieves 88.65 MW, reducing demand MAE by 4.66 MW relative to A1 (approximately 5.0%). Paired Wilcoxon testing reports median demand Δ MAE of +2.81 MW for A1 versus A2 ($p = 0.301$, bootstrap 95% CI [-3.69, 4.93] MW), +3.85 MW for A1 versus A5 ($p = 1.48 \times 10^{-4}$, Bonferroni significant at $\alpha = 0.01$, CI [2.19, 6.90] MW), and -5.43 MW for A1 versus A6 ($p = 5.5 \times 10^{-5}$, CI [-7.17, -2.16] MW).

On the stress task, A2 improves upon A1 (stress MAE 0.0405 versus 0.0499; stress R^2 0.701 versus 0.585), whereas A5 attains the lowest stress MAE in Table V at 0.0340 with stress R^2 of 0.764—marginally above A6 (0.0371 MAE, 0.745 R^2). The ablation evidence shows that graph choice materially affects both tasks, but not uniformly: removing the graph module on the hybrid trunk does not yield a statistically significant demand change, geographical-only adjacency significantly degrades demand, and correlation-only adjacency significantly improves demand while also raising stress R^2 relative to A1. The design implication is that, among the graph variants evaluated in the present study (A2, A5, and A6), correlation-graph conditioning is the only variant that simultaneously delivers statistically supported demand improvement over A1; geographical-only priors should not be retained when demand accuracy is prioritised, and the hybrid graph is not demand-optimal relative to A6 despite its role as the historical reference.

3) *Transformer contribution:* Variant A3 removes the transformer encoder while preserving the hybrid graph and multi-task heads. A3 records demand MAE of 92.64 MW—0.67 MW below A1—and stress MAE of 0.0405 with stress R^2 of 0.701, compared with A1 values of 93.31 MW, 0.0499, and 0.585 respectively. Wilcoxon comparison of A1 versus A3 yields median demand Δ MAE of -1.13 MW, two-sided $p = 0.384$, and bootstrap 95% CI [-2.35, 1.08] MW, which spans zero; the comparison is not significant at the Bonferroni-adjusted threshold $\alpha = 0.01$.

Within the A1–A6 ablation grid, transformer removal on the hybrid topology therefore produces numerically lower demand MAE and improved stress metrics relative to A1, but the demand difference is small and not statistically distinguishable from zero on the test split. The ablation record alone does not establish whether the transformer is redundant on all graph topologies; it demonstrates that, for the hybrid graph reference, A3 and A1 perform similarly on demand MAE. The design implication is that the current ablation does not provide sufficient evidence to recommend transformer removal on hybrid-graph grounds, because the non-significant demand

gap does not justify sacrificing the full encoder without further topology-specific validation.

4) *Multi-task contribution:* Multi-task learning is isolated by variant A4, which disables the OSI head and trains demand only on the hybrid graph. A4 achieves the lowest demand MAE in the ablation suite at 86.89 MW with demand R^2 of 0.731, compared with A1 at 93.31 MW and 0.674. Wilcoxon comparison of A1 versus A4 reports median demand Δ MAE of -5.25 MW, $p = 0.00284$, and bootstrap 95% CI [-10.63, -2.40] MW; A4 improves demand relative to the multi-task reference A1, though the Bonferroni flag for “significantly worse” does not apply because A4 is better, not worse. A4 provides no stress MAE or stress R^2 entries in Table V.

Relative to the final multi-task configuration A6 (S2), A4 retains a 1.76 MW demand advantage (86.89 versus 88.65 MW) but forfeits OSI forecasting entirely. Conversely, A6 improves demand by 4.66 MW over A1 while delivering stress R^2 of 0.745—0.160 above A1—indicating that multi-task training under the correlation graph does not reproduce the demand-only ceiling but does enable joint stress prediction at substantially higher quality than A1. The design implication is that multi-task learning trades approximately 1.76 MW of demand MAE relative to the single-task upper bound in exchange for OSI outputs with R^2 above 0.74; operators requiring only regional demand may prefer A4’s megawatt accuracy, whereas joint load-and-stress planning necessitates a multi-task variant such as A6.

5) *Demand–stress trade-offs across A1–A6:* Fig. 5 orders all six configurations by demand MAE: A4 (86.89 MW), A6 (88.65 MW), A3 (92.64 MW), A1 (93.31 MW), A2 (93.93 MW), and A5 (97.98 MW). Read with Table V, the ranking shows that the lowest demand bar (A4) corresponds to the absence of stress outputs, while the second-lowest (A6) pairs the best multi-task demand score with stress R^2 of 0.745. A5 occupies the highest demand position yet attains the best stress MAE (0.0340) and stress R^2 (0.764) among OSI-producing variants, confirming that stress-optimal and demand-optimal configurations diverge within the frozen suite. A2 and A3 cluster near A1 on demand while recording identical stress MAE (0.0405) and stress R^2 (0.701), suggesting similar joint-task behaviour when either the graph module or transformer is removed on the hybrid trunk.

The benchmark artefacts document these trade-offs as point estimates on 264 test windows; they do not establish causal mechanisms for why A5 excels on stress while lagging on demand. The design implication is that no single ablation row simultaneously minimises demand MAE and maximises both stress metrics: selecting a production configuration requires an explicit priority over joint versus demand-only operation and over demand versus stress emphasis within multi-task candidates.

6) *Final model selection:* Architecture A6 (correlation-only graph, full transformer, multi-task) is designated S2—the final proposed model—based on its position in Table V and Fig. 5. A6 delivers the lowest demand MAE among all multi-task configurations (88.65 MW), with statistically

supported improvement over A1 (median Δ MAE -5.43 MW, $p = 5.5 \times 10^{-5}$, bootstrap 95% CI [-7.17, -2.16] MW). It simultaneously records stress R^2 of 0.745 and stress MAE of 0.0371, exceeding A1 on both stress metrics and matching the S2 row in the external benchmark of Table III. A6 is not demand-optimal within the ablation suite—A4 remains 1.76 MW lower—and is not stress-optimal among multi-task variants—A5 records lower stress MAE and higher stress R^2 at the cost of 9.33 MW additional demand error.

Selection of A6 as the final configuration therefore reflects a joint-task criterion: best multi-task demand accuracy with strong OSI performance and statistically significant improvement over the historical hybrid reference, rather than optimisation of either task in isolation. The design implication for the frozen architecture is adoption of correlation-only graph conditioning with the full transformer trunk and multi-task W20 heads as S2, reflecting the documented demand–stress Pareto trade-off relative to A4 and A5 while meeting the joint forecasting requirement established in Sections 8.1 and 8.3.

E. Error Analysis

1) *Scope and aggregate error magnitudes:* Sections 8.1–8.4 established benchmark and configuration rankings for S2; this subsection characterises the magnitude and structure of remaining forecast errors on the held-out test split rather than restating model superiority. The frozen evaluation record for S2 (variant A6) reports macro demand MAE of 88.65 MW, demand RMSE of 127.29 MW, demand MAPE of 6.55%, and macro demand R^2 of 0.684, together with OSI MAE of 0.0371, OSI RMSE of 0.0473, and OSI R^2 of 0.745 on 264 test windows (Table III; `ablation_raw.json`).

The ratio of demand RMSE to demand MAE is approximately 1.44 (127.29/88.65), indicating that squared-error emphasis falls on a subset of windows or regions with larger absolute deviations than the macro mean. For OSI, the RMSE-to-MAE ratio is approximately 1.27 (0.0473/0.0371), suggesting comparatively less tail amplification relative to demand on the normalised stress scale. Pooled signed-residual statistics—mean bias, skewness, and histogram shape—for S2 are not tabulated in the frozen results package; the planned error-analysis framework defined post-hoc residual protocols, but residual-store generation was not executed for the final checkpoint. Aggregate error magnitudes therefore rest on MAE, RMSE, and R^2 summaries rather than on full distributional diagnostics for S2.

2) *Regional error concentration:* Per-region demand MAE for S2 on the test split ranges from 32.73 MW (Barishal) to 293.98 MW (Dhaka), with Khulna at 88.62 MW, Rajshahi at 76.25 MW, Rangpur at 53.97 MW, Sylhet at 40.62 MW, Mymensingh at 59.22 MW, Chattogram at 76.52 MW, and Cumilla at 75.92 MW (`ablation_raw.json`, variant A6). Dhaka MAE exceeds the next-largest division (Khulna) by more than 205 MW and accounts for the dominant share of the macro numerator despite equal per-division weighting in the macro average.

The same concentration pattern appears for the historical hybrid reference B07 in Table S1 (Dhaka MAE 299.78 MW versus peripheral values generally below 90 MW), indicating that high absolute error in the capital division is a persistent feature of the test period across PF-STGT configurations rather than an artefact isolated to S2. S2 reduces Dhaka MAE by 5.80 MW relative to B07 (293.98 versus 299.78 MW) while retaining a ninefold gap between Dhaka and Barishal MAE. The interpretation is that remaining demand error is structurally concentrated in the largest load centre; the limitation is that per-region signed residuals and box-plot distributions for S2 are not published in the frozen supplementary materials, restricting regional diagnostics to absolute MAE.

3) *Demand versus stress error comparison:* On the common test split, S2 explains approximately 68.4% of macro demand variance ($R^2 = 0.684$) while explaining 74.5% of OSI variance ($R^2 = 0.745$). Absolute demand error is reported in megawatts (macro MAE 88.65 MW), whereas OSI error is reported on the unit interval (MAE 0.0371), so the two tasks are not directly comparable in physical units; the relevant comparison is relative explanatory power and absolute error on each task's native scale.

Demand MAPE of 6.55% indicates that typical regional percentage deviations remain modest at the macro level, but megawatt RMSE of 127.29 MW shows that demand errors on high-load days or in high-load divisions can exceed the mean absolute level. OSI MAE below 0.04 with R^2 of 0.745 indicates that graph-level stress forecasts track the held-out OSI series more closely in variance terms than demand forecasts do, despite demand being the primary optimisation weight in training. Representative test-day case records from Experiment 04 illustrate isolated stress deviations: on 2024-09-08 (high-stress stratum), observed OSI is 0.565 while the S2 prediction is 0.429, whereas on 2024-10-27 (typical-demand stratum) observed OSI is 0.426 and the prediction is 0.411 (`case_studies.md`). The benchmark demonstrates that large single-day OSI gaps can occur on the test split, but four published test case studies are insufficient to characterise systematic temporal or regime-dependent stress bias; that mechanism remains unresolved in the frozen artefacts.

4) *Coalition-level error structure:* Fig. 6 and Fig. 7 present frozen coalition-level summary plots from Experiment 04 for the S2 checkpoint, visualising how forecast deviations associate with grouped input coalitions on the stress task (Fig. 6) and on Dhaka demand (Fig. 7). On the stress task, limitation stack (G8) and calendar/trend (G6) coalitions show the widest spread among the plotted groups; grid aggregates (G7) and national generation scalars (G10) exhibit narrower spread (Fig. 6; frozen Exp04 coalition rankings). On Dhaka demand, calendar/trend (G6), engineered lags and rolling statistics (G4), and national generation scalars (G10) show the widest spread, with regional demand and supply blocks (G1, G2) at intermediate levels (Fig. 7).

The plots indicate that remaining forecast error co-occurs with variability in calendar, limitation, and lagged-demand input coalitions rather than being uniformly distributed across

all feature groups. The repository evidence does not establish whether errors are driven by mis-specification of those coalitions or by unmodelled events outside the input tensor; causal attribution is not resolved by the error summaries alone. The limitation is that these coalition plots summarise association structure on the frozen evaluation sample and do not replace per-window residual traces or signed bias tests for S2.

5) *Temporal behaviour and residual diagnostics:* The pre-defined error-analysis protocol specified temporal, extreme-event, and graph-conditioned error segments, but no executed temporal MAE bins, monthly residual profiles, or pooled residual store for S2 appear in the frozen results package. Temporal error behaviour on the 2024 test period therefore cannot be reported as a segmented time series from repository outputs.

Experiment 04 includes four representative test dates (2024-10-27, 2024-04-28, 2024-12-19, 2024-09-08) spanning typical-demand, high-demand, low-demand, and high-stress strata (`case_studies.md`). These records show day-specific demand totals between 8,866 MW and 16,418 MW and OSI values between 0.344 and 0.565 on the test split, confirming that the held-out year contains diverse operating points, but they do not constitute a systematic temporal error audit. Equivalent pooled signed-residual statistics are not available for S2 in the frozen evaluation record; the historical PF-STGT reference B07 is therefore discussed below for contextual orientation only. Experiment 02A reports pooled residual diagnostics for B07—mean residual -19.58 MW, median -18.97 MW, residual skew -2.998, and pred-to-actual standard-deviation ratio 0.983 with attenuated peak tracking (`residual_analysis.md`, `prediction_distribution_analysis.md`). Those values characterise the prior hybrid-graph configuration and cannot be transferred to S2 without a dedicated residual audit of the final checkpoint. The limitation is that temporal and signed-residual characterisation of S2 remains incomplete in the present evaluation.

6) *Deployment implications:* The frozen error evidence supports deploying S2 as a joint forecaster with macro demand MAE of 88.65 MW and OSI R^2 of 0.745, while treating Dhaka as a separately monitored division: its MAE of 293.98 MW dominates national error budgets despite S2 achieving the lowest audited Dhaka MAE among models with published division tables (Section 8.2). Operators should expect comparatively well-calibrated graph-level stress forecasts on average, with awareness that high-stress test days documented in Experiment 04 can exhibit OSI under-prediction exceeding the macro stress MAE.

Coalition-level summaries in Figs. 6 and 7 suggest that deployment monitoring should include limitation-stack and calendar inputs alongside lagged demand channels, because forecast deviations cluster with variability in those coalitions on the frozen evaluation. Without executed temporal segmentation under the planned error-analysis framework or S2 signed-residual histograms, routine operational guardrails should rely on published macro and per-region MAE/ R^2

thresholds, division-level Dhaka reporting, and periodic stress-case review rather than on repository-supplied uncertainty intervals or automated bias alarms. A limitation of the current evaluation is the absence of complete S2 residual time-series diagnostics and uncertainty bounds; remaining error structure is therefore documented through aggregate metrics, regional concentration, and coalition associations rather than through signed residual traces or forecast confidence intervals on the test split.

F. Explainability Analysis

1) *Scope and attribution methods*: The explainability evaluation applies post-hoc attribution methods [18], [20] to the frozen S2 checkpoint (variant A6, seed 42) without retraining or altering the benchmark metrics reported in Sections 8.1–8.5. Table VII consolidates headline outputs from Experiment 04: grouped GradientSHAP attributions (reported as coalition magnitudes $-\phi-$), coalition-level permutation importance, exported spatial and temporal attention weights, and 24 stratified case studies (20 validation, 4 test). Coalition $-\phi-$ values are obtained through a Captum-compatible integrated-gradients loop using a GradientSHAP approximation with 25 steps and a zero baseline—a SHAP-style attribution pipeline rather than classical exact Shapley enumeration (`shap_summary.md`; `shap_engine.py`).

The frozen record does not include a separate standalone Captum attribution export distinct from this integrated-gradient pipeline; cross-method checks therefore compare gradient-based coalition attributions against permutation importance and OSI component decomposition rather than against an independent Captum ranking. The practical implication is that coalition importance below should be read as model attribution on the frozen validation sample, not as proof of physical causation or operator-ground-truth driver validation.

2) *Coalition-level gradient attribution*: On the global stress task, grouped GradientSHAP magnitudes rank limitation stack (G8) first at $-\phi- = 0.0191$ and calendar/trend (G6) second at $-\phi- = 0.0190$, followed by grid aggregates (G7) at 0.0087 and national generation scalars (G10) at 0.0082 (Table VII; `shap_summary.md`, validation $n = 20$). For Dhaka demand, calendar/trend (G6) records $-\phi- = 162.34$, engineered lags and rolling statistics (G4) record 101.26, and national generation scalars (G10) record 91.44; regional demand and supply blocks (G1, G2) follow at 77.72 and 77.41 respectively, with limitation stack (G8) far lower at 23.15 on this task.

The stress and demand heads therefore emphasise different coalition families: limitation and calendar channels dominate stress attributions, whereas Dhaka demand attributions concentrate on calendar structure, autoregressive lag features, and national generation context. The GradientSHAP pipeline recovers this task-specific separation on the frozen sample. Operators reviewing day-ahead outputs should treat limitation-stack and calendar coalitions as the primary stress attribution channels and calendar/lag coalitions as the primary Dhaka demand channels, while recognising that attribution magnitude

does not establish which inputs physically cause forecast changes.

3) *Spatial node attribution*: Fig. 8 presents the node-level importance heatmap combining mean GradientSHAP node mass and spatial attention on the correlation graph. Dhaka receives the highest mean node attribution mass at 340.36, followed by Rajshahi (110.32), Khulna (108.91), and Myensingh (93.84); peripheral nodes record substantially lower mass, with Rangpur at 1.99 and Chattogram at 6.21 (Table VII; `node_attribution.md`). The Spearman correlation between learned spatial attention and the frozen correlation adjacency is $\rho = 0.422$.

The spatial pattern shows that attribution mass concentrates on divisions with large evening-peak load and strong graph connectivity to Dhaka, consistent with the regional error concentration documented in Section 8.5. Moderate attention–adjacency alignment indicates partial—but not complete—correspondence between learned message passing and the train-estimated correlation prior. The practical implication is that spatial explanations support monitoring Dhaka and its high-mass neighbours jointly, while $\rho = 0.422$ cautions against equating attention patterns with the full correlation graph structure.

4) *Temporal attribution*: Fig. 9 plots mean temporal attention weights α_t across the seven-day input window. Weights are near-uniform, with the largest mean weight at lag t-6 ($\alpha = 0.1622$), followed by t-3 (0.1459) and t-0 (0.1435); remaining lags range from 0.1348 to 0.1405 (`temporal_attribution.md`). Case-study records most frequently list t-6 among the top three temporal lags (Table VII).

The near-uniform profile indicates limited temporal selectivity within the lookback window on the frozen evaluation sample: no single day dominates attention across cases. The model therefore distributes temporal credit broadly rather than concentrating on one lag, an observation that is consistent with the non-significant demand effect of transformer removal on the hybrid reference in Section 8.4, although the present attribution analysis alone cannot establish that relationship. For deployment, temporal attributions suggest that forecasts respond to the full week of context rather than to a single recent day, but they do not identify which calendar periods drive remaining forecast error.

5) *OSI dual-path stress attribution*: Fig. 10 compares grouped GradientSHAP stress attributions with OSI component drivers (reserve margin c_2 versus limitation stack c_3) across the 24 case studies. The dominant stress coalition in case records is G8 (limitation_stack); peak-demand strata sometimes rank G7 (grid_aggregates) highest. Dual-path agreement between the top gradient-attribution coalition and the OSI component driver occurs in 13 of 24 cases (52.2%) (`stress_attribution.md`; `case_studies.md`).

Partial agreement indicates that model attributions and component-level OSI decomposition align in slightly more than half of audited days—for example, high-stress validation cases on 2024-01-19 and 2024-01-06 where G8 matches a

TABLE III
BENCHMARK COMPARISON (TEST SET).

ID	Model	MAE	RMSE	MAPE	R^2	OSI MAE	OSI R^2
S2	Corr.-Only PF-STGT	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B06	T-GCN	257.21	301.06	15.72	-0.483	0.0891	-0.304

TABLE IV
BENCHMARK WILCOXON TESTS (DEMAND MAE; REF. B07).

Comparison	Median Δ MAE	p (two-sided)	Bonferroni sig.
B07 vs B01	-58.98	1.72×10^{-31}	Yes
B07 vs B02	-4.92	0.00135	Yes
B07 vs B06	-160.66	1.48×10^{-40}	Yes

c_3 limitation driver—but diverge in the remainder, including several low-OSI and shedding strata where G8 is the top gradient-attribution coalition while the component driver is c_2 reserve margin. The benchmark demonstrates association between limitation-stack attributions and OSI outputs but does not fully validate gradient-attribution rankings against component physics. Stress explanations should be presented as consistent in a majority of case studies while flagging the 47.8% disagreement rate.

6) *Cross-method validation and limitations:* Permutation importance partially corroborates stress gradient-attribution rankings (Spearman $\rho = 0.366$) but disagrees with Dhaka demand gradient-attribution order ($\rho = -0.564$) (Table VII; `feature_importance.md`). On demand, permutation ranks regional supply block (G2) highest by mean MAE degradation (1.96 MW) while GradientSHAP ranks calendar/trend (G6) highest ($-\phi = 162.34$); engineered lags (G4) show high $-\phi$ magnitude (101.26) but low permutation degradation (0.013 MW). On stress, limitation stack (G8) leads both gradient-attribution and permutation tables, though permutation deltas are numerically small (top stress permutation $\Delta \approx 3.8 \times 10^{-5}$).

Because no independent Captum-only attribution table is published separately from the integrated-gradient pipeline, gradient-attribution versus Captum agreement cannot be assessed directly; the relevant disagreement is between gradient-based attributions and permutation-based importance, especially on demand. Global GradientSHAP summaries are computed on validation windows ($n = 20$); case-study dual-path statistics cover 24 stratified days with only four on the test split. Attribution outputs describe model behaviour on the frozen sample and must not be interpreted as causal proof, complete OSI driver validation, or interchangeable rankings across methods. Practical use of S2 explanations should combine coalition gradient attributions, spatial attention (Fig. 8), temporal weights (Fig. 9), and component comparison (Fig. 10) with explicit reporting of partial cross-method and dual-path agreement. Taken together, the explainability artefacts provide complementary transparency into model behaviour rather than independent validation of the underlying physical mechanisms.

TABLE V
ABLATION STUDY RESULTS (TEST SET).

ID	Variant	MAE	R^2	OSI MAE	OSI R^2
A4	Single-Task	86.89	0.731	—	—
A6 (S2)	Corr. Graph	88.65	0.684	0.0371	0.745
A1	PF-STGT W20	93.31	0.674	0.0499	0.585
A5	Geo Graph	97.98	0.554	0.0340	0.764

TABLE VI
ARCHITECTURE COMPARISON (S1–S4).

ID	Model	MAE	R^2	OSI R^2
S2	Corr.-Only PF-STGT	88.65	0.684	0.745
S3	No-Transformer	92.64	0.671	0.701
S1	PF-STGT W20	93.31	0.674	0.585
S4	Corr. + No-Trans.	114.63	0.362	0.747

TABLE VII
EXPLAINABILITY SUMMARY (S2).

Metric	Value
Top stress coalition (G8 $ \phi $)	0.0191
Top Dhaka demand coalition (G6 $ \phi $)	162.34
Attention–adjacency Spearman ρ	0.422
SHAP–permutation ρ (demand)	-0.564
OSI driver agreement (cases)	52.2%

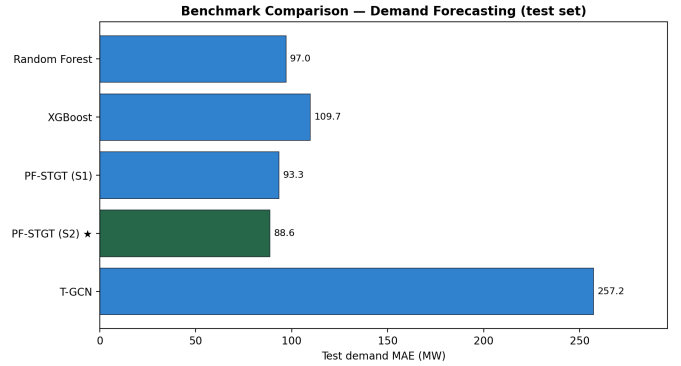


Fig. 4. Test-set macro demand MAE benchmark comparison.

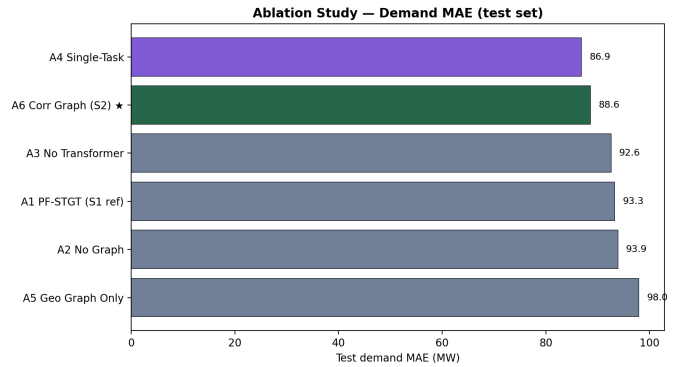


Fig. 5. Ablation study demand MAE on the test set.

VI. DISCUSSION

A. Principal Findings

1) *Overall contribution:* The completed study delivers an empirically frozen Correlation-Aware Multi-Task Forecasting

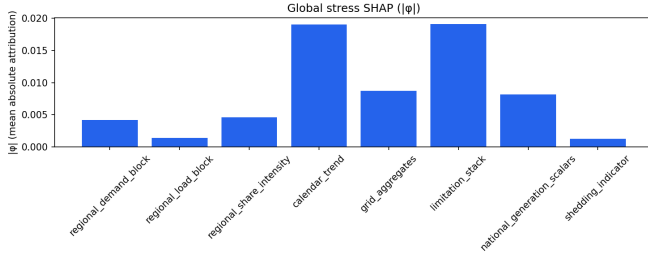


Fig. 6. Grouped SHAP attributions for stress (global).

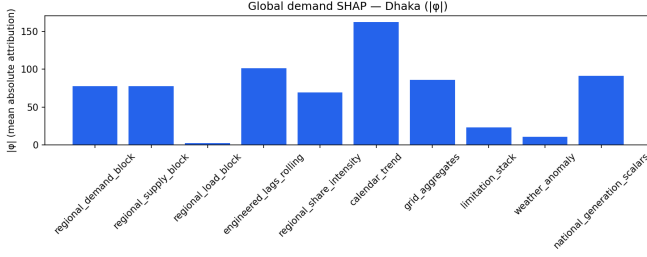


Fig. 7. Grouped SHAP attributions for Dhaka demand.

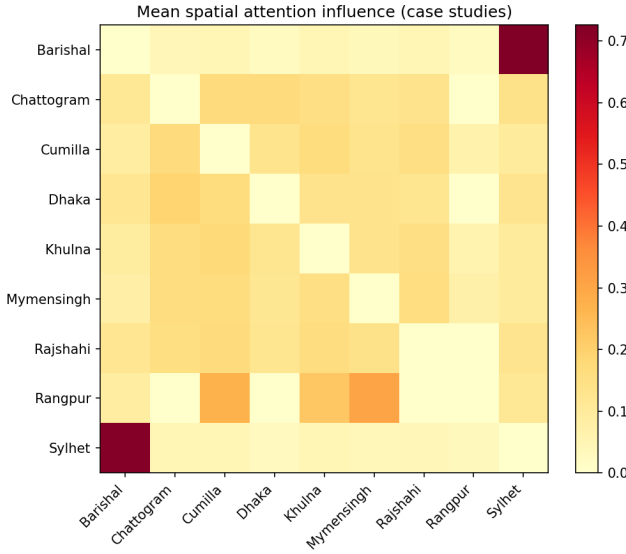


Fig. 8. Node-level attribution heatmap.

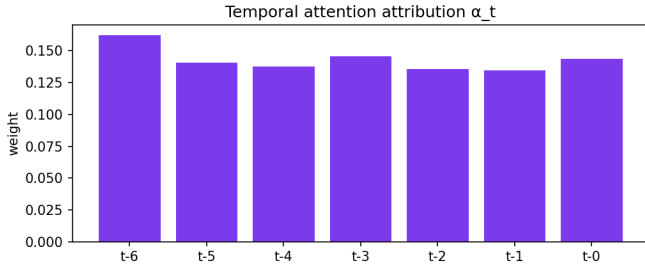


Fig. 9. Temporal attention weights across the lookback window.

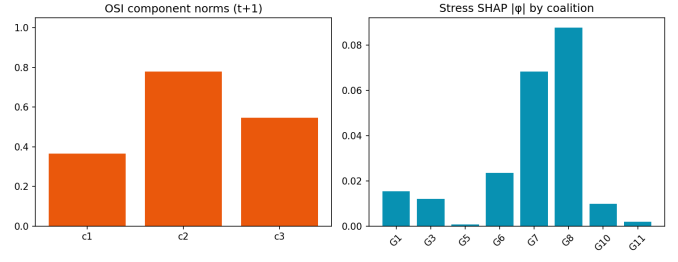


Fig. 10. Dual-path stress attribution vs. OSI component drivers.

Framework that forecasts next-day regional evening-peak demand and a graph-level Operational Stress Index from a single spatio-temporal graph-transformer model [7], [17] on the Bangladesh national power system [1]. The principal finding is not superiority on every isolated metric, but the demonstration that one architecture can supply operationally coupled day-ahead load and stress visibility from shared representations without reverting to separate single-task pipelines.

That joint capability matters because regional dispatch and system-pressure monitoring are ordinarily served by different models or post-hoc indicators. The frozen evaluation programme—spanning external benchmarks, configuration ablations, architecture selection, error characterisation, and post-hoc attribution—shows that the final correlation-graph PF-STGT configuration (S2) was selected through evidence rather than design intent alone. Scientifically, the work positions graph-transformer multi-task learning as a viable modelling paradigm for developing-economy grids where limitation dynamics and inter-regional load coupling coexist in daily operational records. Practically, operators gain a single checkpoint that supports paired load-and-stress dashboards instead of reconciling disconnected forecasts after the fact.

2) *Interaction between demand forecasting and stress prediction:* A central finding is that demand and OSI forecasting are complementary tasks within one model rather than redundant projections of the same signal. Stress-capable baselines in the benchmark suite show that classical ensembles can attach auxiliary stress outputs, yet their stress explanations of variance remain materially weaker than those of the proposed multi-task graph-transformer, while shallow recurrent and graph-convolution architectures fail to produce informative OSI forecasts at all. Within the configuration suite, lower demand error does not automatically imply better stress fidelity: geographically specialised and demand-only variants illustrate a genuine demand–stress trade-off space rather than a single Pareto-optimal point.

The interpretation is that shared encoding captures calendar, limitation, and grid-context structure relevant to both targets, while task-specific heads preserve distinct output semantics. Multi-task training [14], [19] under the repaired W20 protocol restores stress learnability that earlier configurations lacked, indicating that task balancing—not architecture alone—gates joint success. Scientifically, the study supports multi-task formulation as a principled response to coupled operational

questions rather than as an incidental add-on to demand modelling. For day-ahead operations, the practical value lies in synchronised update cycles: the same input window yields regional megawatt trajectories and a national stress scalar suitable for joint review in control-room workflows.

3) *Architecture selection and empirical simplification:*

Another principal finding is that evidence-driven simplification of the graph prior improves the frozen model relative to the historical hybrid geographical–correlation design. Correlation-only adjacency emerges as the strongest graph variant among those evaluated for joint demand and stress performance, whereas geographical-only conditioning degrades demand accuracy and removing the graph module on the hybrid trunk does not yield a statistically distinguishable gain. The architecture-selection record further shows that simplification is not uniformly beneficial: eliminating the temporal transformer on the correlation graph results in a marked degradation of demand performance, even though transformer removal on the hybrid reference is statistically indistinguishable on demand error.

The interpretation is that predictive inter-region coupling in this dataset is better encoded by train-estimated correlation structure than by static geographical proximity alone, and that the temporal branch appears to play a critical role once the graph prior is correlation-based. Scientifically, the result cautions against importing composite graph designs from other domains without ablative justification on the target system. Practically, deployment should adopt the correlation-graph, full-transformer, multi-task configuration frozen as S2 rather than lighter variants that appear attractive on isolated ablation rows.

4) *Regional robustness and error structure:*

Regional analysis reveals a robust but heterogeneous forecaster: peripheral divisions achieve comparatively low absolute error, while the national load centre dominates the megawatt error budget across model families. The proposed model improves the capital division relative to earlier PF-STGT and strong classical references without eliminating the structural concentration of residual mass in the largest division. Macro-averaged scores therefore understate planning significance for the principal load hub even when they support a favourable national summary.

The interpretation is that regional robustness in this study means stable multi-division service with predictable error geography, not uniform accuracy everywhere. Error concentration in the highest-variability division persists across architectures, suggesting that remaining demand difficulty is tied to scale and volatility of the dominant node rather than to a failure of graph conditioning alone. Scientifically, the finding reinforces the need to report division-level diagnostics alongside macro metrics in multi-node national forecasts. Practically, operators should treat Dhaka as a separately reported diagnostic in deployment monitoring while relying on the model for consistent peripheral and mid-scale division service.

5) *Explainability contribution:*

Post-hoc attribution on the frozen S2 checkpoint shows that stress and demand predic-

tions draw on interpretable coalition families—limitation and calendar structure for OSI, calendar and autoregressive lag channels for the principal load division—while spatial mass concentrates on high-load nodes with moderate alignment between learned attention and the correlation adjacency. Dual-path comparison between gradient attributions and OSI component drivers achieves partial, not universal, agreement across stratified case studies, and demand attributions disagree with permutation-based importance rankings.

The interpretation is that explainability artefacts offer complementary transparency into model behaviour rather than independent validation of physical mechanisms. Coherent global and spatial attributions support operator trust in the sense of traceable inputs, but partial cross-method and dual-path disagreement forbids treating attribution plots as ground-truth causal maps. Scientifically, the study demonstrates multi-method post-hoc analysis in a national-scale graph-transformer forecasting setting, with explicit reporting of discordance where methods diverge. Practically, limitation-stack and calendar coalitions should headline stress explanation dashboards, while demand explanations should present gradient-based and permutation-based views separately rather than collapsing them into a single ranked list.

6) *Scientific significance:*

Taken together, the principal findings establish that correlation-graph PF-STGT multi-task learning can jointly model regional demand and operational stress on a chronologically held-out year of Bangladesh grid data with statistically supported improvement over the historical hybrid reference on the primary demand objective and markedly stronger stress explanatory power within the frozen benchmark suite. The study contributes an evidence-locked modelling chain—from task repair and benchmark comparison through ablation, architecture freeze, error geography, and attribution—that is reproducible without rerunning training.

Scientifically, the work extends spatio-temporal deep forecasting beyond load-only benchmarks to coupled stress indices derived from operational limitation dynamics, and it shows that graph priors and multi-task objectives must be empirically tuned rather than assumed from domain intuition. For power-system practice in data-rich developing economies, the implication is that day-ahead joint load-and-stress forecasting is feasible as a single-model service when graph structure, task weighting, and division-level monitoring are co-designed with the evaluation evidence reported in Section 8.

B. *Comparison with Existing Literature*

1) *Graph-based spatial forecasting:*

The present study finds that train-estimated correlation adjacency outperforms geographical-only and hybrid geographical–correlation priors for joint regional demand and OSI forecasting on the Bangladesh national grid, while a shallow spatio-temporal graph-convolution baseline demonstrates substantially lower predictive performance than the graph-transformer family in the frozen benchmark suite. Prior graph-neural work in power systems—represented in the reviewed corpus by a small set of high-relevance spatio-temporal GNN and graph-attention stud-

ies [8], [9], [26]—typically couples nodes through geographic proximity, charging-network topology, or integrated-energy-system structure on US and European case studies rather than on developing-economy division-level operational records.

That contrast aligns with a recurring limitation in the literature: graph designs validated on residential, microgrid, or AMI-centric networks do not transfer automatically to a nine-division national system where inter-region demand coupling is strong and limitation dynamics are nationally co-visible. Recent spatio-temporal graph-attention and multi-graph convolution studies [15], [16], [25] demonstrate strong performance within their native topologies but seldom co-design graph priors with limitation-stack covariates or national stress targets. The scientific implication is that data-driven correlation graphs can encode predictive spatial structure where static geography alone is insufficient, extending the graph-forecasting line beyond topology imported from infrastructure maps. A remaining gap is the absence of external replication on other South Asian or similarly structured national grids; the correlation-graph advantage documented here is evidence-backed for Bangladesh but not established as a universal graph-construction rule.

2) *Transformer-based temporal modelling*: The frozen architecture-selection record shows that temporal transformer capacity appears critical when the graph prior is correlation-based, whereas transformer removal on the historical hybrid reference is not statistically distinguishable on demand error—a pattern not commonly reported in transformer forecasting papers that evaluate a single backbone on a fixed graph. Transformer-based load forecasting [22], [23] constitutes a larger but still minority share of the reviewed literature, often applied to residential sequences, EV charging, or multivariate energy series without multi-node national stress heads.

Compared with those studies, the present work does not claim a novel temporal operator; rather, it shows that transformer contribution is conditional on graph topology in a multi-task national setting. Near-uniform temporal attention weights further suggest limited lag selectivity, which diverges from narratives that attribute transformer gains primarily to sharp recency focus. Scientifically, the finding cautions that transformer value in spatio-temporal energy forecasting should be reported together with graph context and ablation evidence, not assumed from single-configuration benchmarks. The remaining gap is whether alternative temporal encoders could match transformer performance on correlation graphs with lower complexity—an question the present simplification study begins but does not exhaust.

3) *Multi-task demand and operational stress learning*: A principal outcome of this study is that regional demand and graph-level OSI can be learned jointly from shared representations when task weighting is explicitly repaired, yielding a favourable joint configuration relative to single-task and stress-weak baselines, while still exhibiting a documented demand–stress trade-off against a demand-only upper bound. The reviewed literature rarely couples continuous multi-node demand forecasting with operational-stress-style targets in

one model: multi-task entries in the corpus are sparse, and the shedding cluster predominantly addresses control, optimisation, or under-frequency response rather than day-ahead forecasting of coupled load and pressure indices.

Relative to hierarchical multi-energy multi-task proposals [19], [24], the present formulation shifts emphasis from sectoral energy vectors to limitation-aware operational stress alongside evening-peak demand—closer to operational-stress gaps in the literature than to classical load-only multi-task extensions. Classical ensembles in the frozen benchmark can emit auxiliary stress outputs, which superficially resembles multi-target learning, but the literature seldom evaluates whether such add-on heads share representations meaningfully with spatial graph encoders. The scientific implication is that multi-task learning in energy systems is shown to be viable within the evaluated Bangladesh forecasting setting when tasks are operationally related yet statistically non-redundant, but not when treated as a default regulariser without loss calibration. The remaining gap is limited cross-dataset evidence on task-interference regimes: the W20 repair and ablation trade-offs are documented for one national timeline, not for a family of stress definitions or shedding formulations.

4) *Explainability in graph-temporal forecasting*: Post-hoc attribution on the frozen model shows coherent coalition-level and spatial patterns—limitation and calendar structure for stress, calendar and lag channels for the principal load division—with partial dual-path agreement against OSI component drivers and discordance between gradient-based and permutation-based demand rankings. Explainability appears in only a small fraction of the reviewed corpus; where present, it often serves microgrid shedding control, distillation, or generic interpretable ML frameworks rather than multi-node graph-transformer forecasts with operator-facing dual-task outputs.

Compared with SHAP-centric shedding studies [11], [18] and review papers that catalogue XAI methods [12] for scalar load series, the present work embeds explainability inside a national multi-task graph-transformer pipeline and reports method disagreement explicitly rather than treating a single attribution map as validation. This sits closer to the small set of interpretable smart-grid forecasting papers [13] than to the majority of deep load models in the corpus that report no post-hoc attribution at all. The scientific implication is that explainability for graph-temporal energy models must be multi-method and task-specific: stress attributions partially align with engineered OSI components, but demand attributions are not interchangeable across techniques. The remaining gap is the lack of prospective operator studies linking attribution outputs to decision quality; transparency is demonstrated on frozen artefacts, not human-in-the-loop efficacy.

5) *Bangladesh smart-grid context and evaluation discipline*: The Bangladesh case [1] and spatial load-shedding literature [10] provide contextual anchors for interpreting the frozen evaluation record.

Taken together, the empirical programme addresses a geographic and contextual void in the reviewed literature: no collected comparator used Bangladesh division-level smart-

grid daily data, and developing-economy national grids remain under-represented in graph-transformer forecasting benchmarks dominated by US and European residential or microgrid settings. The study further departs from much of the shedding literature by forecasting day-ahead regional load and a derived operational stress index for planning visibility rather than optimising real-time load-shedding actions.

Relative to papers with weakly documented splits or metadata-sparse conference reporting, the frozen evaluation chain—chronological hold-out testing, benchmark and ablation suites, inferential testing on the primary demand metric, and locked artefacts—responds to reproducibility concerns identified in the gap analysis without claiming superiority over unbenchmarked external systems. The scientific implication is that rigorous internal benchmarking on a previously unreported national case addresses an important gap identified in the literature even when architectural components are incremental at the operator level. The remaining gap is external validity: conclusions are anchored to the Bangladesh timeline and feature regime; transfer to other developing grids requires new datasets, stress definitions, and independent replication rather than architectural export alone.

C. Practical Implications

1) *Day-ahead operational planning:* The completed evaluation shows that the frozen correlation-graph PF-STGT model (S2) delivers the strongest joint demand-and-stress credentials in the benchmark suite on a chronologically held-out test year, with statistically supported improvement over the historical hybrid reference on the primary demand objective and comparatively informative graph-level OSI outputs. For day-ahead operational planning, this may support a single forecasting service that updates regional evening-peak load trajectories and a national stress scalar from the same input window rather than reconciling separate demand and stress indicators after inference.

Operationally, planners could review paired forecasts in a common daily cycle aligned with the one-day horizon and seven-day lookback used in evaluation, which is consistent with existing evening-peak planning rhythms on the national system. Deployment would require adopting the frozen S2 checkpoint and the associated feature and graph artefacts without altering the locked train-only preprocessing rules that governed the reported scores, together with version control that preserves MD5-locked data dependencies referenced in the final results package. A practical caution is that the study does not quantify downstream dispatch, reserve, or shedding outcomes; forecast accuracy improvements documented in Section 8 are planning diagnostics only and should not be read as demonstrated reductions in operational risk or economic cost.

2) *Regional monitoring:* Regional disaggregation shows that absolute forecast error concentrates in the national load centre while peripheral divisions exhibit comparatively low megawatt deviations, a pattern that persists across model families in the frozen record. Macro-averaged national summaries

therefore may understate planning sensitivity in the largest division and may overstate uniform national improvement when division-level offsets are present.

Operationally, regional monitoring should treat the capital division as a separately reported diagnostic alongside equal-weight macro dashboards, with supplementary division tables consulted when prioritising corridors or load centres with the largest residuals. Deployment could facilitate a two-tier reporting layout: a national summary for routine briefings and division-level panels for Dhaka, Khulna, and other high-residual nodes identified in the ablation outputs. A practical caution is that per-region signed-residual distributions and variance diagnostics for S2 are incomplete in the published supplementary materials; monitoring protocols should track division MAE and, where available, audited R^2 from reference models rather than assuming uniform error character everywhere.

3) *Joint demand and OSI forecasting:* The study demonstrates that demand-only and stress-optimal configurations occupy different positions in a documented trade-off space: a demand-only variant attains lower megawatt error at the cost of forfeiting OSI outputs, while stress-favouring graph choices can improve OSI fidelity at higher demand error. The selected multi-task configuration is the empirically preferred joint option within the frozen suite, not simultaneous optimum on every task.

Operationally, organisations that require only regional load forecasts may consider a demand-only configuration, whereas control centres that need synchronised load and stress visibility may favour the multi-task S2 pipeline evaluated here. Deployment of the joint model may facilitate paired dashboards for next-day regional demand and graph-level OSI without maintaining two independent training and versioning streams. A practical caution is that OSI performance claims rest primarily on point estimates without stress-specific confidence intervals or hypothesis tests; joint monitoring should treat OSI forecasts as complementary stress indicators to be reviewed alongside demand accuracy, not as validated triggers for automatic control actions.

4) *Explainability for operator support:* Post-hoc attribution on the frozen model indicates that limitation-stack and calendar coalitions dominate stress explanations, while calendar, lag, and national-generation coalitions feature prominently for the principal load division, with partial agreement between gradient attributions and OSI component drivers across audited case studies. Spatial attribution further concentrates on high-load nodes with moderate alignment between learned attention and the correlation adjacency. Explainability artefacts therefore may support operator review by surfacing which input coalitions co-vary with forecasts, not by proving which physical mechanisms caused outcomes.

Operationally, stress explanation views could foreground limitation and calendar channels, and demand explanation views for the capital region could present calendar and lag coalitions alongside spatial mass on high-load nodes. Deployment could assist trust-building by displaying gradient-

based and permutation-based demand attributions separately, consistent with documented disagreement between methods, and by flagging case-study strata where dual-path OSI agreement is absent. A practical caution is that attributions are model-local and partially discordant with component-level OSI decomposition; operators should not treat explanation plots as ground-truth causal maps or as substitutes for engineering judgement on high-stress days where case records show larger single-day OSI gaps.

5) *Deployment considerations*: The frozen evaluation programme—benchmark comparison, ablation and architecture selection, error geography, and attribution analysis on locked artefacts—provides a reproducible evidence base for considering S2, but it does not constitute a field deployment trial. Integration would require the chronological feature store, correlation adjacency, W20 multi-task checkpoint, and inference pipeline consistent with the experimental setup, together with monitoring against published macro and division-level accuracy thresholds and periodic retraining policy aligned with the chronological split discipline used in evaluation.

Operationally, shadow-mode forecasting alongside incumbent tools before any operational cutover is a recommended deployment strategy that the present study did not evaluate; such a prudent rollout may support division-level and stress-case review built into acceptance checks. Deployment planning should budget for ongoing monitoring of limitation and calendar inputs highlighted in error and attribution analyses, and for separate Dhaka reporting given persistent error concentration. Operational guidance should distinguish macro national summaries from division diagnostics and from explanation outputs that use different attribution methods. A practical caution is that absence of complete S2 residual time-series diagnostics and forecast uncertainty bounds in the current evaluation limits automated bias alarms; human oversight remains necessary until extended error-analysis protocols are executed and operator-facing validation is conducted beyond the frozen test record.

D. Limitations

1) *Dataset scope*: The empirical evidence is bounded by a single chronologically split daily record of the Bangladesh national power system: nine administrative divisions, a seven-day input window, a one-day evening-peak forecast horizon, and a feature regime comprising nine node-level and seventeen global channels derived from the frozen preprocessing pipeline. That design supports reproducible within-dataset comparison but does not span alternative temporal resolutions, intra-day dispatch horizons, or feature sets outside the locked parquet artefacts.

The limitation arises because model selection, graph construction, and OSI definition were calibrated on one national timeline (train through mid-2023, validation through early 2024, test on 264 windows through late 2024) with warm-up rules and MD5-locked inputs fixed before publication freeze. Interpretation of accuracy rankings, ablation outcomes, and attribution patterns therefore applies to this feature geome-

try and calendar span rather than to arbitrary grid telemetry schemas or longer institutional archives. Conclusions on joint demand-and-stress forecasting are scoped to the documented Bangladesh feature store and chronological split; they do not establish performance under different data vintages, missing-data regimes, or revised stress formulations without re-execution of the frozen evaluation chain.

2) *Bangladesh-only evaluation*: All reported benchmarks, ablations, error characterisations, and explainability outputs derive from the Bangladesh national grid case alone; no independent replication on other South Asian or developing-economy systems is included in the frozen record. The study addresses a geographic void in the reviewed literature but remains a single-country evidence base.

National grids differ in topology, limitation reporting, shedding practice, and divisional aggregation; correlation-graph priors and limitation-stack coalitions that perform well here were estimated from Bangladesh operational records and may not transfer where coupling structure or stress semantics differ. Interpretation of graph-topology findings, multi-task trade-offs, and regional error geography is therefore country-specific: external validity is limited to systems with comparable divisional structure, limitation visibility, and daily planning cadence. Claims about graph-transformer multi-task feasibility extend to the evaluated Bangladesh setting; they do not constitute cross-national generalisation without new datasets and independent validation.

3) *Benchmark scope*: Comparative evidence is confined to the frozen internal suite—seven reference models (B01–B07) plus the final proposed configuration S2—and six ablation variants (A1–A6) drawn from the same PF-STGT family and training protocol. No external state-of-the-art leaderboard, proprietary operator model, or real-time control benchmark participates in the reported rankings.

Inferential testing is defined primarily on per-window macro demand MAE: Wilcoxon comparisons and Bonferroni-adjusted significance apply to demand error, with direct S2-versus-random-forest testing absent from the statistical record. Classical ensembles exhibit favourable macro R^2 without leading on megawatt MAE, and a demand-only ablation (A4) attains lower demand error than the selected multi-task configuration, documenting an explicit demand–stress trade-off within the suite rather than universal superiority on every metric. Interpretation of “leading” performance is therefore bounded to the frozen benchmark and ablation tables on the primary demand objective and reported OSI point estimates; it does not assert dominance over unbenchmarked external systems or simultaneous optimality on demand and stress across all configuration rows.

4) *Uncertainty estimation*: The evaluation reports point forecasts and aggregate error metrics—MAE, RMSE, R^2 , and MAPE on demand; MAE and R^2 on OSI—without prediction intervals, ensemble spread, conformal bounds, or other uncertainty quantification on the held-out test split. Bootstrap confidence intervals exist for selected paired demand MAE

contrasts but do not cover OSI residuals or per-division S2 forecasts.

Stress-specific hypothesis tests, multiple-comparison correction on OSI metrics, and pooled signed-residual diagnostics for the final checkpoint were not executed in the frozen results package; temporal error segmentation defined in the error-analysis protocol likewise remains unreported for S2. Interpretation of OSI gains and regional error structure therefore rests on point estimates and association summaries rather than on distributional or interval-based risk statements. Supported demand improvements over the historical hybrid reference retain inferential backing; stress accuracy, bias characterisation, and forecast confidence for deployment monitoring must be read with the narrower inferential and diagnostic coverage documented in Section 8.

5) *Explainability limits*: Post-hoc attribution was applied to the frozen S2 checkpoint using grouped integrated gradients, permutation importance, attention export, and stratified case studies; these methods describe model behaviour on the evaluation sample and do not establish physical causation or operator-ground-truth driver validation. Global gradient attributions for demand were computed on twenty validation windows; dual-path OSI comparison draws on twenty-four stratified days, of which only four lie on the test split.

Cross-method agreement is partial: OSI component drivers align with the top gradient-attribution coalition in 52.2% of audited case studies, demand gradient and permutation rankings disagree (Spearman $\rho = -0.564$), and spatial attention correlates moderately with the correlation adjacency ($\rho = 0.422$). Near-uniform temporal attention weights further limit claims of sharp lag selectivity from attribution outputs alone. Interpretation of explainability artefacts is limited to transparency into the frozen model’s input sensitivities; they do not validate OSI physics, confirm feature rankings across methods, or substitute for engineering judgement on high-stress days where dual-path records diverge. Explainability conclusions are scoped to the documented multi-method discordance and sample sizes in Table VII.

6) *Deployment limits*: The completed programme is an offline, chronologically held-out evaluation on locked artefacts; it does not include field deployment, operator-in-the-loop trials, shadow-mode operation, or quantification of downstream dispatch, reserve, shedding, or economic outcomes. Production integration would depend on reproducing the frozen feature store, correlation adjacency, W20 multi-task checkpoint, and inference pipeline under the same train-only preprocessing rules that governed reported scores.

Single-seed training (seed 42), incomplete S2 residual time-series stores, and absence of forecast uncertainty bounds constrain operational readiness assessment to accuracy diagnostics published in the frozen tables rather than to validated runtime monitoring or cutover evidence. Interpretation of practical implications in Section 9.3 is advisory: deployment strategies discussed there are recommendations grounded in evaluation gaps, not outcomes measured in the present study. The boundary of the reported evidence is reproducible forecasting per-

formance and post-hoc analysis on the Bangladesh test year; operational adoption, reliability effects, and cost consequences lie outside the scope of the frozen evaluation record.

E. Future Research Directions

1) *External validation*: The completed study establishes that correlation-graph PF-STGT multi-task learning can jointly forecast regional evening-peak demand and a graph-level Operational Stress Index on a chronologically held-out year of Bangladesh division-level data, with statistically supported demand improvement over the historical hybrid reference within the frozen benchmark suite. That evidence is internally rigorous but geographically singular.

Developing-economy national grids differ in divisional aggregation, limitation reporting, and inter-regional coupling; the correlation prior and OSI semantics calibrated here may not transfer without re-estimation on independent records. Future studies could replicate the frozen evaluation protocol—chronological splitting, joint demand–stress targets, ablation of graph priors, and multi-method attribution—on comparable South Asian or similarly structured national systems with documented operational stress indicators. Further investigation may compare whether correlation-graph advantage persists where geographical coupling is weaker or limitation dynamics are reported differently.

Such external validation would separate architecture-specific gains from country-specific feature geometry, strengthening or qualifying the generalisability of data-driven graph priors for national-scale energy forecasting. The scientific value lies in converting a single-country demonstration into cross-context evidence on when joint load-and-stress graph-transformer forecasting remains viable outside the Bangladesh timeline.

2) *Richer graph construction*: Ablation evidence shows that correlation-only adjacency outperforms geographical-only and hybrid priors for joint demand and stress performance on this dataset, while a geographical graph variant attains superior OSI point estimates at materially higher demand error, documenting a demand–stress trade-off rather than a single optimal topology. Transformer capacity further appears conditional on graph choice: removal on the correlation graph degrades demand performance sharply, whereas hybrid-topology simplification does not yield statistically distinguishable demand change.

The frozen suite evaluates static correlation ($\tau = 0.65$), geographical, and hybrid adjacencies but does not explore dynamic, multi-relational, or jointly learned graph structures that could encode time-varying coupling or fuse infrastructure topology with data-driven edges. An important direction is comparative evaluation of adaptive correlation graphs, multi-graph attention, and physics-informed edge constraints under the same multi-task objective and chronological discipline used here. Further investigation may examine whether richer priors can narrow the documented gap between stress-favouring and demand-favouring configurations without reverting to the hybrid reference design.

Advancing graph construction beyond the frozen ablation grid would clarify whether correlation-graph simplification represents a robust national-grid design principle or a Bangladesh-specific optimum. The scientific value is a more complete map of graph–task interactions in multi-node operational forecasting, informing prior selection where limitation dynamics and load coupling co-evolve.

3) *Uncertainty-aware forecasting*: Demand-side inferential support in the frozen record includes paired Wilcoxon tests and bootstrap confidence intervals on macro MAE contrasts, whereas OSI performance is reported through point estimates without stress-specific hypothesis tests, prediction intervals, or pooled signed-residual diagnostics for the final checkpoint. Planners reviewing joint forecasts therefore lack distributional statements on per-division error and stress uncertainty on the held-out split.

Probabilistic forecasting—through deep ensembles, quantile heads, conformal calibration, or Bayesian approximations—could extend the present point-forecast pipeline to supply interval-valued demand and OSI outputs aligned with the same chronological evaluation framework. Complementary work may execute the predefined residual and temporal error protocols on the frozen S2 checkpoint, including signed-bias characterisation and regime-conditioned error segments, together with inferential testing on OSI residuals where sample design permits. Further investigation may assess whether uncertainty bands improve discrimination on high-stress test days where case records show larger single-day OSI gaps.

Uncertainty-aware extensions would shift interpretation from accuracy rankings alone toward risk-aware day-ahead planning metrics. The scientific value is coupling the demonstrated joint forecasting capability with statistically auditable confidence statements suitable for operational monitoring and automated bias detection.

4) *Explainability validation*: Post-hoc attribution on the frozen model surfaces coherent coalition-level and spatial patterns—limitation and calendar structure for stress, calendar and lag channels for the principal load division—with explicit documentation of partial dual-path agreement against OSI component drivers and discordance between gradient-based and permutation-based demand rankings. Transparency is demonstrated on frozen artefacts rather than through prospective validation with domain experts.

Attribution samples are modest relative to the full test split, and explainability methods disagree on demand feature order; neither outcome establishes that explanation outputs improve operator decisions or align with ground-truth causal structure on high-stress days. Future studies could expand attribution coverage across held-out test windows, standardise multi-method reporting templates for dual-task dashboards, and conduct controlled operator studies that measure whether discordant explanation views affect forecast trust or review behaviour. Further investigation may compare attribution outputs against independent engineering assessments of limitation events and reserve conditions on stratified stress cases.

Prospective explainability validation would bridge post-hoc model transparency and operational decision quality. The scientific value is empirical evidence on which attribution families are stable, task-specific, and useful for human review in national graph-transformer forecasting settings where method disagreement is already documented.

5) *Deployment studies*: The frozen evaluation programme delivers reproducible offline accuracy diagnostics, architecture selection, and post-hoc analysis on locked data artefacts, but it does not measure runtime integration, shadow-mode operation, data-drift response, or downstream planning outcomes. Practical implications in Section 9.3 therefore remain conditional on evidence not collected in the present study.

Translating benchmark performance into operational utility requires field-oriented study designs that preserve the chronological and preprocessing discipline of the frozen pipeline while exposing the model to production data feeds, incumbent forecast comparisons, and division-level monitoring workflows. Future studies could implement shadow-mode forecasting alongside existing tools, document retraining triggers under extending timelines, and record operator review of paired demand–OSI dashboards under realistic planning cycles. Further investigation may quantify whether published macro and division-level error thresholds remain stable when feature distributions shift beyond the 2024 test year.

Deployment-oriented research would connect forecasting accuracy to observable operational review practices without substituting for the present offline evidence base. The scientific value is closing the evidentiary loop between reproducible model evaluation and documented behaviour under conditions that approximate control-centre use, while maintaining the boundary between planning diagnostics and unmeasured reliability or economic outcomes.

VII. CONCLUSION

Day-ahead electricity planning in developing-economy national grids requires synchronised visibility over regional load and system-wide operational pressure, yet forecasting practice often treats demand prediction and stress monitoring as separate pipelines. This study addressed that coupling by investigating whether a single spatio-temporal deep learning framework can jointly forecast next-day regional evening-peak demand and a graph-level Operational Stress Index from shared historical operational records on the Bangladesh national power system [1]. The research objective was to design, empirically validate, and reproducibly freeze a correlation-aware multi-task forecasting framework under chronological evaluation discipline, with explicit comparison against classical, recurrent, and graph-convolution baselines and systematic ablation of graph priors, architectural components, and task formulation.

The proposed Correlation-Aware Multi-Task Forecasting Framework is implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The model ingests a seven-day lookback of node-level and national context features over nine administrative divisions, conditions representations

on a train-estimated correlation graph, and decodes parallel regional demand and system-wide stress outputs through shared spatio-temporal encoding and task-specific heads. A repaired multi-task training protocol balances demand and stress objectives so that joint learning remains stable rather than collapsing to a single-task solution. The final frozen configuration—correlation-graph PF-STGT with full temporal capacity and multi-task heads—was selected through an evidence-locked programme spanning benchmark comparison, configuration ablation, architecture simplification, error geography, and post-hoc attribution, rather than through design intent alone.

The completed evaluation establishes several substantive contributions. First, the study demonstrates that graph-transformer multi-task learning can deliver operationally coupled day-ahead load and stress forecasts from one checkpoint on a chronologically held-out national timeline, extending spatio-temporal energy forecasting beyond load-only benchmarks to limitation-aware stress indices derived from daily grid records. Second, empirical ablation shows that predictive inter-regional coupling in this setting is better served by data-driven correlation adjacency than by geographical or hybrid graph priors imported without dataset-specific justification, while temporal encoding remains consequential once the correlation prior is adopted. Third, the work documents a genuine demand–stress trade-off space within the configuration suite: demand-only training attains a lower megawatt error ceiling, whereas multi-task training enables informative stress outputs at a modest demand offset relative to that ceiling, positioning the frozen model as the preferred joint rather than single-task optimum. Fourth, regional diagnostics reveal a heterogeneous but stable error geography in which peripheral divisions are comparatively well served while the national load centre dominates absolute residual mass—a pattern that persists across model families and informs how national summaries should be read. Fifth, multi-method post-hoc attribution supplies coalition-level, spatial, and temporal transparency for both tasks, with explicit reporting of partial cross-method and dual-path discordance rather than treating any single explanation map as physical validation.

Scientifically, the study contributes a reproducible modelling chain on a previously unreported Bangladesh division-level smart-grid case, responding to geographic and methodological gaps identified in the reviewed literature on graph-based load forecasting, transformer temporal modelling, multi-task energy learning, and explainability in power systems [4], [12]. The evidence supports correlation-graph PF-STGT multi-task learning as a viable paradigm where limitation dynamics and inter-regional load coupling coexist in operational data, and it cautions that graph priors, task weighting, and architectural simplification must be empirically tuned rather than assumed from domain intuition or transferred from residential and microgrid settings. The contribution is bounded to the evaluated national context and frozen internal benchmark suite; it advances internal rigour and case specificity rather than claiming universal architectural dominance or external state-of-the-art superiority.

For power-system practice, the principal implication is feasibility: a single forecasting service may supply paired regional demand trajectories and a national stress scalar for day-ahead review from the same input window, which could assist planning workflows that presently reconcile disconnected indicators after inference. Operators may treat macro summaries as routine briefings while reporting the principal load centre separately, reviewing limitation and calendar input channels highlighted in error and attribution analyses, and presenting gradient-based and permutation-based explanation views distinctly where methods disagree. These implications are planning diagnostics grounded in offline accuracy and transparency evidence; they do not constitute demonstrated improvements in grid reliability, economic outcome, or deployment success.

In sum, this work delivers an empirically frozen Correlation-Aware Multi-Task Forecasting Framework that jointly models regional demand and operational stress on Bangladesh national grid data through correlation-graph PF-STGT learning, supported by a transparent evaluation record from task formulation through architecture selection and post-hoc analysis. The study shows that coupled day-ahead load-and-stress forecasting is achievable as a single-model service when graph structure, multi-task calibration, division-level monitoring, and explanation reporting are co-designed with the evidence collected. The conclusions stand as a disciplined contribution to spatio-temporal energy forecasting in developing-economy settings—informative for researchers and practitioners who require reproducible joint forecasts and interpretable model behaviour, and appropriately qualified as a completed offline evaluation rather than an operational field trial.

APPENDIX A SUPPLEMENTARY MATERIALS

This appendix consolidates technical material supporting the main manuscript. All entries reference artefacts locked under publication freeze `publication-freeze-2026-06-25` (git commit `dda83f1d9201d55ad8daf6b4cc0456569a84b6aa`). Numeric values in tables below are copied from frozen CSV/JSON outputs; no experiments were rerun for this document.

A. Complete benchmark tables

1) *External benchmark comparison (test set)*: Macro demand metrics are averaged over nine administrative divisions. **S2** is the final proposed model (Experiment 03, variant A6); **B07** is the historical PF-STGT W20 hybrid reference (S1). Stress columns apply only to models that emit graph-level OSI forecasts.

ID	Model	Demand MAE (MW)	Demand RMSE (MW)	Demand MAPE (%)	Demand R^2	Stress MAE	Stress R^2
S2	Correlation-Only PF-STGT (final)	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid (S1 ref.)	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B01	Linear Regression	247.79	597.01	17.32	0.170	0.1074	-1.824
B04	LSTM	237.03	278.67	14.35	-0.242	0.0861	-0.191
B05	GRU	233.48	274.39	14.13	-0.201	0.0863	-0.214
B06	T-GCN	257.21	301.06	15.72	-0.483	0.0891	-0.304

Source: experiments/experiment_02_benchmark_models/benchmark_results.csv (B01–B07); experiments/experiment_03_ablation_studies/ablation_results.csv (S2 = A6).

2) *Ablation study (test set)*: Reference configuration **A1** = S1 (hybrid graph, multi-task W20). **A6** = S2 (final architecture).

ID	Variant	Graph	Multi-task	Demand MAE (MW)	Demand R^2	Stress MAE	Stress R^2
A4	Single-Task	hybrid	No	86.89	0.731	—	—
A6	Correlation Graph Only (S2)	corr	Yes	88.65	0.684	0.0371	0.745
A3	No Transformer	hybrid	Yes	92.64	0.671	0.0405	0.701
A1	PF-STGT W20 (S1 ref.)	hybrid	Yes	93.31	0.674	0.0499	0.585
A2	No Graph	hybrid	Yes	93.93	0.701	0.0405	0.701
A5	Geographical Graph Only	geo	Yes	97.98	0.554	0.0340	0.764

Source: experiments/experiment_03_ablation_studies/ablation_results.csv, ablation_raw.json.

3) *Architecture simplification (test set)*:

ID	Model	Graph	Transformer	Demand MAE (MW)	Demand R^2	Stress R^2	Active params
S2	Correlation-Only PF-STGT	corr	Yes	88.65	0.684	0.745	749,058
S3	No-Transformer PF-STGT	hybrid	No	92.64	0.671	0.701	451,202
S1	PF-STGT W20 (ref.)	hybrid	Yes	93.31	0.674	0.585	749,058
S4	Corr + No-Transformer	corr	No	114.63	0.362	0.747	451,202

Source: experiments/experiment_03B_architecture_simplification/simplification_results.csv.

4) *Per-region demand MAE — S2 (test set)*: Per-division MAE for the final model is available in the frozen ablation record. Per-division R^2 for S2 is **not** tabulated in the frozen repository (macro demand R^2 only).

Region	S2 demand MAE (MW)
Barishal	32.73
Chattogram	76.52
Cumilla	75.92
Dhaka	293.98
Khulna	88.62
Mymensingh	59.22
Rajshahi	76.25
Rangpur	53.97
Sylhet	40.62

Source: experiments/experiment_03_ablation_studies/ablation_raw.json (variant A6, per_region_mae).

5) *Per-region verification — B07 and B02 (test set)*: Division-level MAE and R^2 for the historical hybrid reference (B07) and random forest (B02) were recomputed in Experiment 02A.

Region	B07 MAE (MW)	B07 R^2	B02 MAE (MW)	B02 R^2
Barishal	30.85	0.4808	32.07	0.6936
Chattogram	75.04	0.6362	79.97	0.5941
Cumilla	71.83	0.7930	73.63	0.7688
Dhaka	299.78	0.5744	311.59	0.6666
Khulna	118.72	0.7353	92.06	0.7835
Mymensingh	58.74	0.7584	85.85	0.6126
Rajshahi	85.70	0.8270	83.13	0.8274
Rangpur	59.77	0.6928	68.41	0.6264
Sylhet	39.35	0.5707	46.54	0.6171

Source: experiments/experiment_02A_classical_benchmark_verification/aggregation_audit.md.

Unavailable in frozen repository: Per-region demand MAE and R^2 tables for B01, B04, B05, and B06; per-region R^2 for S2; materialised per-window prediction arrays for S2

(Experiment 02A stores predictions for B07, B02, and B03 only).

B. Additional statistical summaries

1) *Benchmark Wilcoxon tests (demand MAE)*: Paired Wilcoxon signed-rank tests [34] on per-window macro demand MAE (test set, $n = 264$). Reference model: **B07 (S1)**. Bonferroni-adjusted [35] $\alpha = 0.0083$ (six comparisons).

Comparison	Median Δ MAE (MW)	p (two-sided)	Cohen's d	Bonferroni sig.	Bootstrap 95% CI
B07 vs B01	-58.98	1.72×10^{-31}	-0.491	Yes	[-193.11, -120.58]
B07 vs B02	-4.92	0.00135	-0.077	Yes	[-8.87, 2.62]
B07 vs B03	-14.13	6.68×10^{-12}	-0.298	Yes	[-22.76, -9.55]
B07 vs B04	-134.42	2.33×10^{-39}	-1.224	Yes	[-158.49, -129.48]
B07 vs B05	-128.94	1.92×10^{-39}	-1.219	Yes	[-154.54, -126.62]
B07 vs B06	-160.66	1.48×10^{-40}	-1.296	Yes	[-179.85, -148.85]

Source: experiments/experiment_02_benchmark_models/statistical_significance.md.

2) *Ablation Wilcoxon tests (demand MAE)*: Paired Wilcoxon signed-rank tests [34] vs reference **A1**. Bonferroni-adjusted [35] $\alpha = 0.01$ (five comparisons).

Comparison	Median Δ MAE (MW)	p (two-sided)	Bonferroni sig. (worse)	Bootstrap 95% CI
A1 vs A6 (S2)	-5.43	5.5×10^{-5}	No (A6 better)	[-7.17, -2.16]
A1 vs A4	-5.25	0.00284	No (A4 better demand)	[-10.63, -2.40]
A1 vs A5	+3.85	1.48×10^{-4}	Yes (A5 worse demand)	[2.19, 6.90]
A1 vs A2	+2.81	0.301	No	[-3.69, 4.93]
A1 vs A3	-1.13	0.384	No	[-2.35, 1.08]

Source: experiments/experiment_03_ablation_studies/statistical_significance.md.

3) *S2 vs S1 and R^2 aggregation note*:

Item	Value / note	Source
S2 vs S1 (A6 vs A1)	Median Δ MAE = -5.43 MW; $p = 5.5 \times 10^{-5}$; bootstrap 95% CI [-7.17, -2.16]	Exp03 statistical summary
B02/B03 reported R^2 in Exp02	Equals pooled R^2 , not macro R^2	aggregation_audit.md
B07 reported R^2	Equals macro R^2 (mean of nine per-region R^2)	metric_verification.md
Unified macro R^2 (recomputed)	B02: 0.6878; B07: 0.6743; B03: 0.5902	Exp02A

Unavailable in frozen repository: Paired Wilcoxon test of S2 vs B02 (random forest); Wilcoxon tests, bootstrap confidence intervals, or Bonferroni-corrected hypothesis tests on OSI MAE or OSI R^2 ; pooled signed-residual statistics or residual time-series store for S2; temporal error-segmentation outputs defined in the Phase 14 error-analysis protocol.

C. Hyperparameter configuration

Frozen training configuration for the final model (S2 = A6, seed 42). Authority: Experiment 01B W20 repair protocol and A6 checkpoint config.

Setting	Value
Architecture	PFSTGT + GraphVariant.CORR ($\tau = 0.65$)
Tasks	Multi-task: regional demand + graph-level OSI
Loss	$L = \text{Huber}(\text{demand})/100 + \lambda_2 \cdot \text{MSE}(\text{OSI})$
λ_2 (stress weight)	20.0
Optimiser	Adam, $\text{lr} = 5 \times 10^{-4}$, weight decay = 10^{-4}
Batch size	32
Max epochs	200 (early stopping epoch 69 for A6)
Early stopping	Patience 15; score = $0.7 \cdot \text{val_demand_MAE}/100 + 0.3 \cdot \text{val_stress_MAE}$
Scheduler	ReduceLROnPlateau on validation demand MAE; factor 0.5, patience 5
Gradient clipping	Max norm 1.0
Seed	42
Parameters	749,058
S2 checkpoint	experiments/experiment_03_ablation_studies/checkpoints/A6/seed_42/best.pt
S1 reference checkpoint	experiments/experiment_01B_multitask_optimization_repair/checkpoints/W20/B07/seed_42/best.pt

Supporting documents: experiments/experiment_01B_multitask_optimization_repair/best_configuration.md, experiments/architecture_freeze_revision/final_model_specification.md.

Unavailable in frozen repository: Multi-seed training sweeps (single seed 42 only); forecast uncertainty or probabilistic head configuration.

D. Feature dictionary

1) Model input tensors:

Tensor	Shape (per batch)	Channels	Description
Node features	(T=7; N=9; F _{node} =9)	9 per division	Contemporaneous demand/supply/load; lags and rolling mean; regional share and load intensity
Global features	(T=7; F _{global} =17)	17	Calendar/trend; grid aggregates; limitation stack; weather; generation scalars; shedding indicator

Leakage policy: Same-day OSI(t) is excluded from model inputs when forecasting OSI(t+1). The engineered stress composite used in target construction is withheld from the global input tensor at inference.

2) Node-level channels ($F_n = 9$ per division):

Channel group	Contents
Power balance (contemporaneous)	Processed evening-peak demand, supply, and load (MW-scale, standardised)
Autoregressive temporal	Demand lag (1 obs.), demand lag (7 obs.), load lag (1 obs.), 7-observation demand rolling mean
Regional derived	Division demand share; load intensity (load / max(demand, ϵ))

3) Global-level channels ($F_g = 17$):

Channel group	Contents
Calendar and trend	Day-of-year sin/cos, monotonic trend index, inter-observation gap days, holiday-category ordinal
Grid aggregates	Total regional demand, total regional load, generation reserve
Operational	Total operational limitation (aggregated), any-division shedding indicator (binary)
Weather	Temperature anomaly (month-relative, train-fitted means)
Limitation stack	Gas, coal, water, and maintenance limitation scalars (preprocessed)
National generation	Two preprocessed generation-endpoint scalars

4) Coalition grouping (explainability): Eleven coalitions support grouped attribution in Experiment 04:

ID	Coalition name	Scope	Primary contents
G1	regional_demand_block	Node	Contemporaneous division demand
G2	regional_supply_block	Node	Contemporaneous division supply
G3	regional_load_block	Node	Contemporaneous division load
G4	engineered_lags_rolling	Node	Demand lags and rolling mean
G5	regional_share_intensity	Node	Demand share and load intensity
G6	calendar_trend	Global	Cyclical calendar, trend, gap, holiday summary
G7	grid_aggregates	Global	Total demand, total load, generation reserve
G8	limitation_stack	Global	Fuel, infrastructure, maintenance limitations
G9	weather_anomaly	Global	Temperature anomaly
G10	national_generation_scalars	Global	Generation-endpoint scalars
G11	shedding_indicator	Global	Binary any-division shedding flag

Design registry: results/phases/phase_05A_feature_blueprint/feature_inventory.csv (Phase 05A blueprint; implementation matches Section 6.4 of the manuscript).

Unavailable in frozen repository: A standalone column-name manifest exported from data/features/*_features.parquet is not included as a separate supplementary CSV; feature columns are defined in Section 6.4 and the coalition table above.

E. Graph construction details

1) Final model — correlation graph (S2):

Property	Value
Construction rule	Undirected edge (i,j) retained when train-split Pearson $\rho_{ij} \geq \tau$ on regional evening-peak demand
Threshold τ	0.65
Train rows for ρ	1,295 daily records (training partition only)
Retained pairs	33 of 36 undirected pairs (91.7% density)
Edge weight	$w_{ij} = \rho_{ij}$ (zero diagonal; no self-loops)
Normalisation	Row-normalised adjacency A (outgoing weights sum to unity)
Temporal behaviour	Static: estimated once from training data; fixed for validation and test
Runtime variant	GraphVariant.CORR in src/models/pf_sgt.py

Attention bias: Derived deterministically from A (log-scaled positive entries; masked absent edges) for additive spatial masking in the graph-transformer branch.

2) Alternative graphs evaluated in ablations:

Variant	Rule	Undirected edges	Density	Notes
Geographical	Border-adjacent divisions; uniform binary weights, row-normalised	21	58.3%	Frozen geographic prior in graph/adjacency_matrices.csv (MD5-locked)
Hybrid (S1)	Edge if geographic neighbour or $\rho_{ij} \geq 0.85$; weight = ρ_{ij}	24	66.7%	Historical reference graph for A1/B07
Correlation (S2)	Edge if $\rho_{ij} \geq 0.65$; weight = ρ_{ij}	33	91.7%	Selected final prior

Design-phase report: graphs/graph_construction_report.md documents the hybrid-graph design rationale (Phase 08); the publication freeze selects correlation-only adjacency for S2 per experiments/architecture_freeze_revision/Final_Architecture_Decision.md.

Unavailable in frozen repository: A persisted CSV of the full correlation adjacency matrix (built at runtime from training demand series); dynamic or rolling-window graph variants.

F. Additional explainability outputs

Experiment 04 post-hoc analysis on frozen S2 checkpoint (A6/seed_42/best.pt).

1) Summary reports:

Report	Path
XAI summary	experiments/experiment_04_explainability_analysis/xai_summary.md
SHAP summary	experiments/experiment_04_explainability_analysis/shap_summary.md
Permutation importance	experiments/experiment_04_explainability_analysis/feature_importance.md
Node attribution	experiments/experiment_04_explainability_analysis/node_attribution.md
Temporal attribution	experiments/experiment_04_explainability_analysis/temporal_attribution.md
Stress dual-path attribution	experiments/experiment_04_explainability_analysis/stress_attribution.md
Case studies (24 dates)	experiments/experiment_04_explainability_analysis/case_studies.md
Machine-readable metrics	experiments/experiment_04_explainability_analysis/xai_metrics.json

2) Tabular and figure artefacts:

Label	Content	Path
Table S2	Global grouped SHAP (stress)	results/explainability/shap/global_stress.csv
Table S2	Global grouped SHAP (Dhaka demand)	results/explainability/shap/global_demand_dhaka.csv
Table S3	Permutation Δ MAE (demand)	results/explainability/permutation/demand_importance.csv
Table S3	Permutation Δ (stress)	results/explainability/permutation/stress_importance.csv
Table S4	Case-study records	results/explainability/case_studies/date/ (24 date folders)
Figure S1	Signed stress SHAP bar	experiments/experiment_04_explainability_analysis/figures/figure_shap_bar_stress.png
Figures 3-9	Manuscript explainability figures	manuscript/overleaf/figures/ (copies from Exp04)

3) Cross-method headline metrics (S2):

Metric	Value
Attention–adjacency Spearman ρ	0.422
SHAP–permutation Spearman (demand)	-0.564
SHAP–permutation Spearman (stress)	0.366
OSI driver agreement (case studies)	52.2% (13/24)
Global demand SHAP sample	n = 20 validation windows
Case-study dates on test split	4 of 24

Unavailable in frozen repository: Independent Captum attribution export separate from the integrated-gradient SHAP-style pipeline; prospective operator-in-the-loop evaluation records; explainability outputs for ablation variants other than S2.

G. Reproducibility checklist

1) Freeze metadata:

Item	Value
Freeze date	2026-06-25
Designated tag	publication-freeze-2026-06-25
Git commit (freeze HEAD)	dda83f1d9201d55ad8daf6b4cc0456569a84b6aa
Final model ID	S2 — Correlation-Aware Multi-Task Forecasting Framework
Freeze authority	paper/publication_freeze/Publication_Asset_Freeze.md

2) MD5-locked data artefacts:

File	MD5
data/interim/bangladesh_smartgrid_clean.parquet	4255024d735a91a4b53b2edee203d0ca
data/features/train_features.parquet	b8b3bda95d0fd6cc65f4910d85a98e16
graphs/adjacency_matrix.csv	dac67ac3a827d00a4b61ea9400e75686

3) Chronological split policy:

Split	Date range	Valid windows (T = 7)
Train	2019-11-21 → 2023-06-15	1,281
Validation	2023-06-16 → 2024-03-19	263
Test	2024-03-20 → 2024-12-30	264

Model selection on validation only; test partition held out until final evaluation. Warm-up skip: seven rows per split.

4) Experiment completeness (frozen):

Experiment	Primary artefact	Status
01 / 01A / 01B	W20 protocol + S1 checkpoint	Complete
02	benchmark_results.csv	Complete
02A	Verification reports + B07/B02/B03 predictions	Complete
03	ablation_results.csv + A6 checkpoint	Complete
03A / 03B	Investigation reports + simplification CSV	Complete
04	xai_metrics.json + figures + CSV tree	Complete

5) *Artefacts not included in the frozen repository:* The following items referenced in the manuscript or design documentation are **not** present as executed outputs in the frozen record:

1. Per-region demand R^2 table for S2. 2. Pooled signed-residual diagnostics, residual histograms, or residual time-series store for S2. 3. Materialised per-window prediction arrays for S2 (B07/B02/B03 only in Exp02A). 4. OSI-specific Wilcoxon tests, bootstrap intervals, or multiple-comparison correction on stress metrics. 5. Direct paired inferential test of S2 vs random forest (B02). 6. Temporal error-segmentation bins or monthly residual profiles for S2 (Phase 14 protocol defined but not executed for the final checkpoint). 7. Forecast prediction intervals, ensemble spread, or conformal uncertainty bounds. 8. Persisted correlation-adjacency matrix file (runtime construction from training demand). 9. Multi-seed result tables (seed 42 only). 10. Field deployment, shadow-mode, or operator-in-the-loop trial records.

Reproduction of reported results should use the checkpoint, data MD5 hashes, split policy, and experiment scripts referenced in paper/publication_freeze/frozen_results_inventory.md without modifying frozen CSV, JSON, or figure artefacts.

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